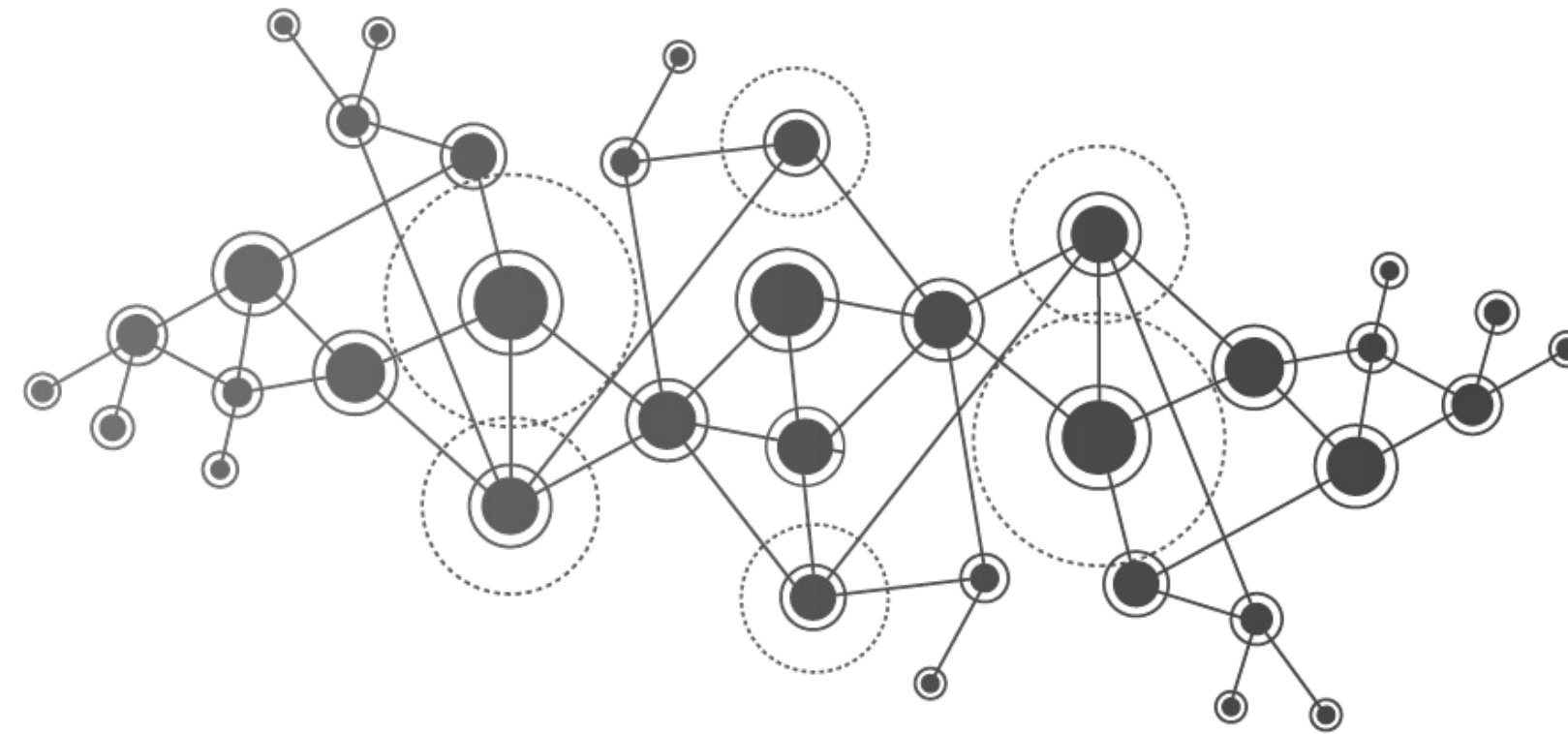




UNIMORE

UNIVERSITÀ DEGLI STUDI DI
MODENA E REGGIO EMILIA



Intelligent Internet of Things

Lecture 1- Internet of Things Overview

Prof. Marco Picone

A.A 2026/2027

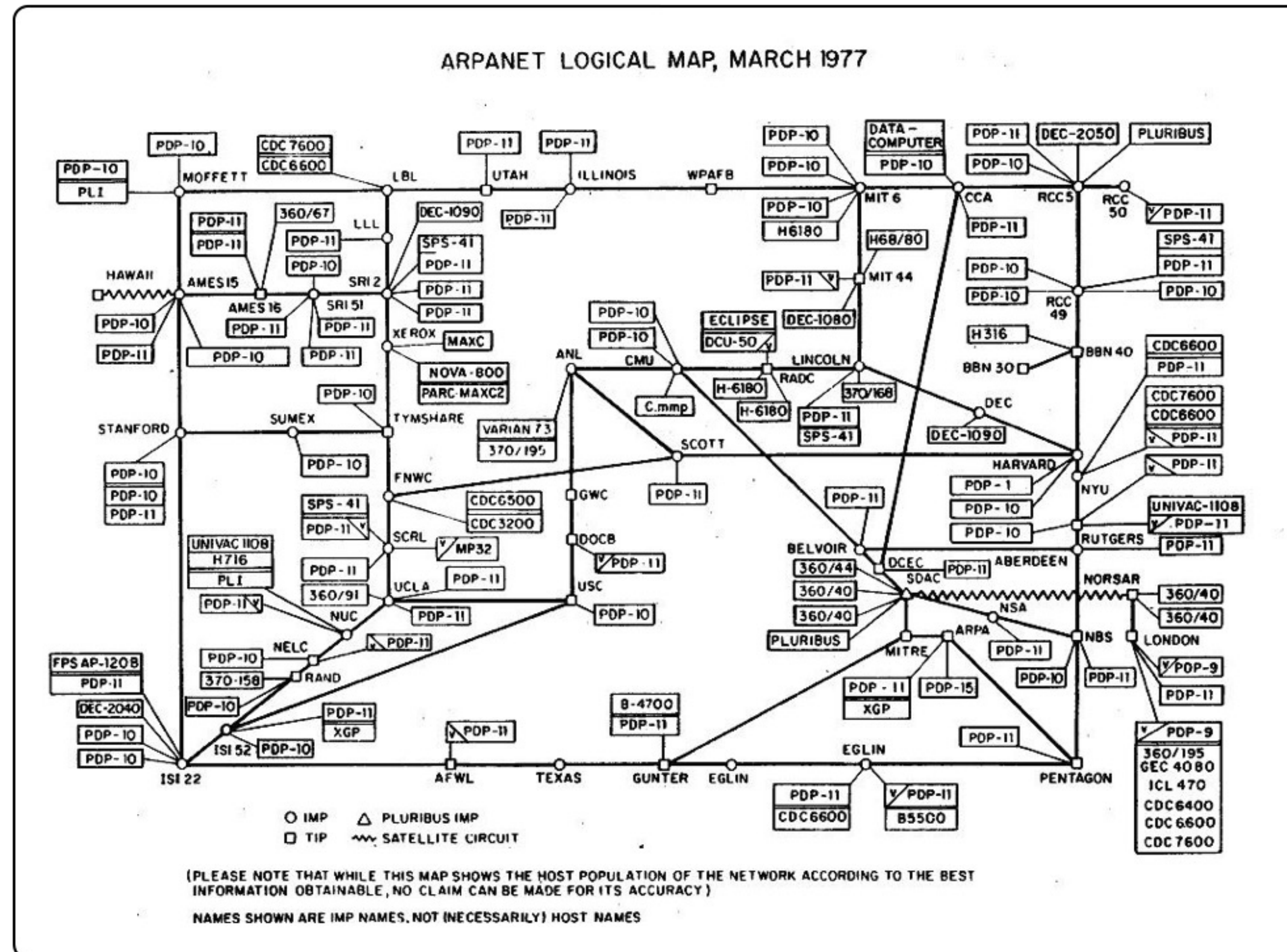
Internet of Things Overview

- Introduction
- Internet of Things Definition & Vision
- From WSN and M2M to IoT
- IoT Characteristics
- IoT Protocol Stack
- IoT Applications

Where do we started ?

Since from the beginning of the Internet, there was the desire to connect more “things” to it.

Source: ARPANET
Completion Report:
January 4, 1978.

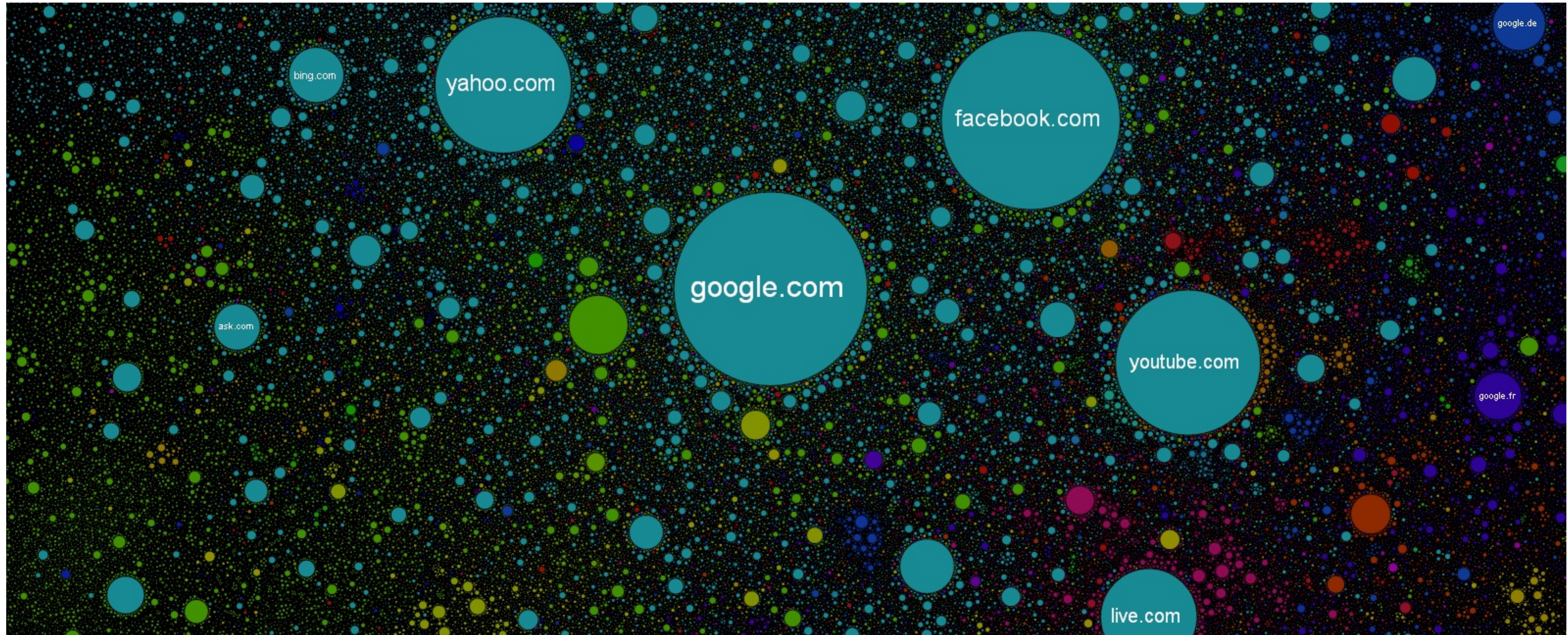


<http://www.youtube.com/watch?v=vuiBTJZfeo8>

<http://www.lk.cs.ucla.edu/index.html>

Where are we today ?

The internet now connects billions devices anywhere!



The first “IoT” Connected Devices

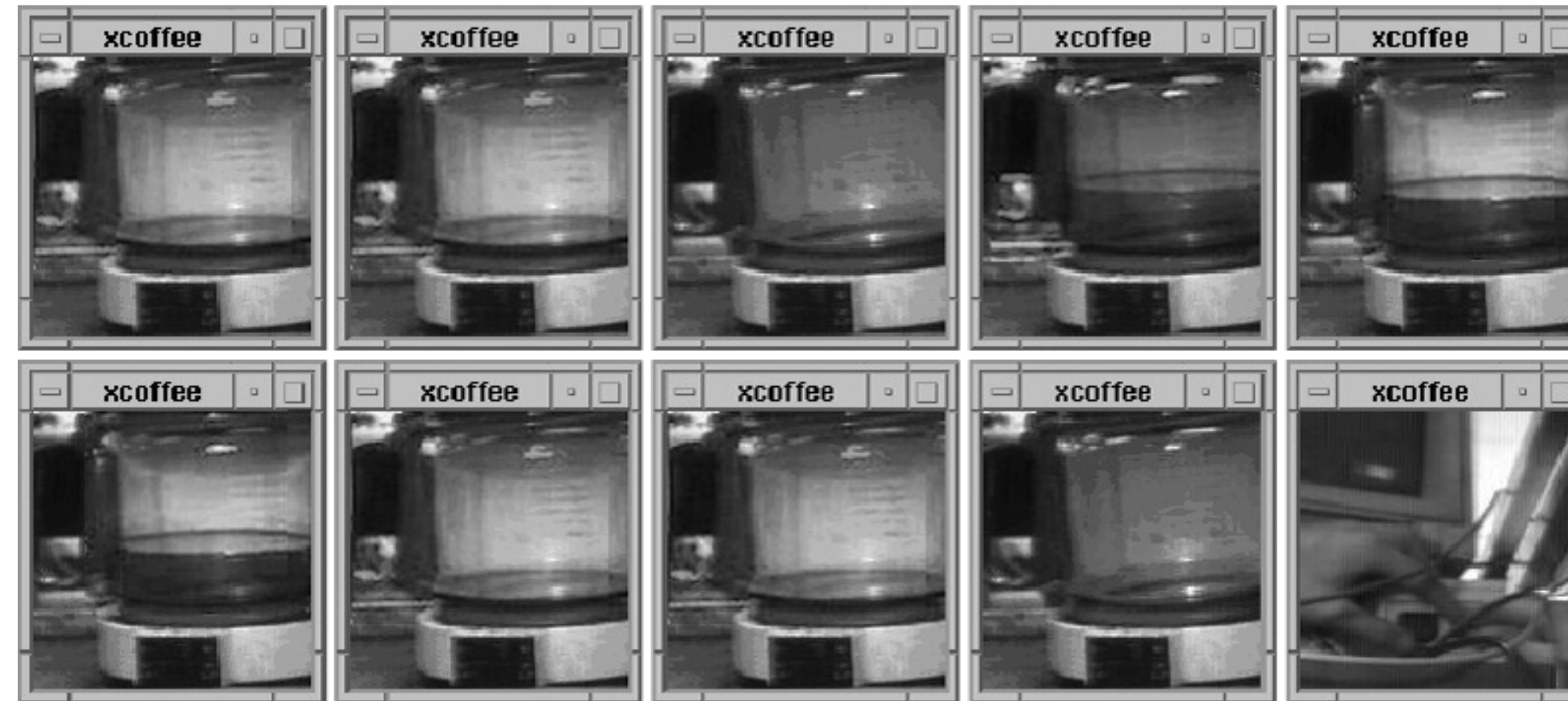


“The students of Carnegie Mellon University invent the ARPANET-connected coke machine. Four students from the School of Computer Science department installed the switches in the coke machine. They counted how many coke bottles had remained in each row and for how long. If the loaded bottle spends a long time in the machine, it is labeled “cold”. All this data was then remotely available to customers via a finger interface. This experiment had inspired a lot of inventors all over the world to create their own connected appliances.”

CSE3002 Team Project History of Internet of Things (IoT) Devices and Their Influence on Cyber Security Team F Christopher Byars Dylan Leonard Brian Patino.
- <https://slideplayer.com/slide/14090495/>

<https://hqsoftwarelab.com/blog/the-history-of-iot-a-comprehensive-timeline-of-major-events-infographic/>

The first “IoT” Connected Devices



“The XCoffee is created at the Computer Laboratory of the University of Cambridge. Two academics who worked at the Trojan Room in the laboratory had a video-frame grabber attached to one of the computers. They decided to connect a camera controlled by a server application that was observing the coffee pot and taking pictures of it several times a minute. Then the pictures were sent to the client application, to keep the user updated on the information about whether the pot was filled.”

<https://www.cl.cam.ac.uk/coffee/qsf/coffee.html>

<https://odannyboy.medium.com/the-first-things-on-the-internet-of-things-5d817d45ac9c>

<https://hqsoftwarelab.com/blog/the-history-of-iot-a-comprehensive-timeline-of-major-events-infographic/>

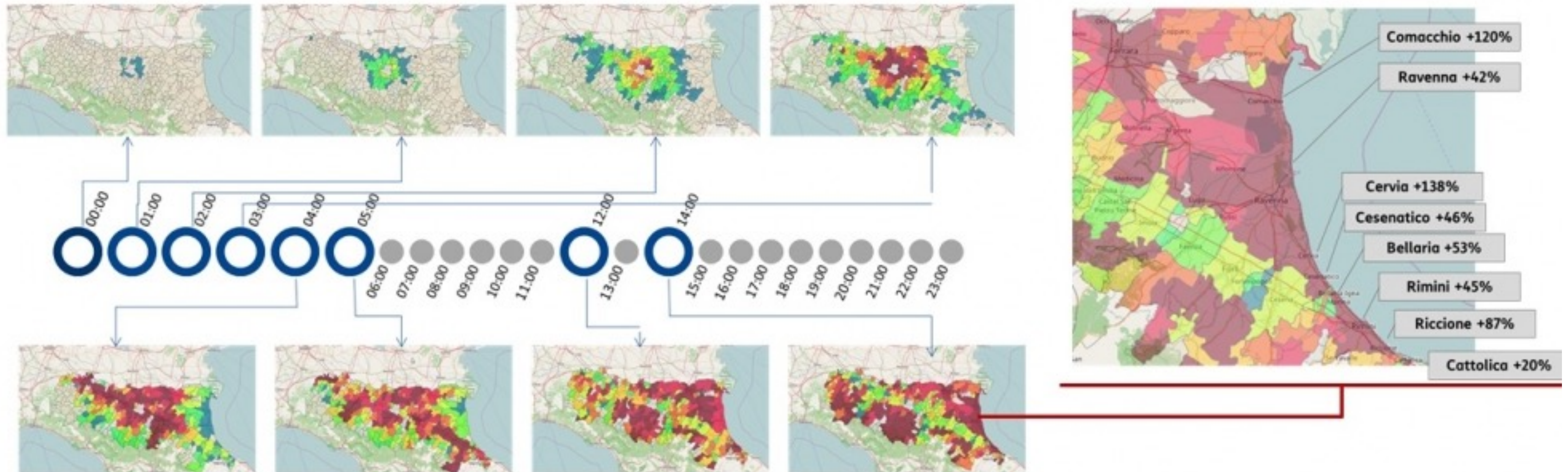
The Mobile Centric World



“What a difference 8 years makes:
St. Peter’s Square in 2005 and
yesterday (2013)” (Facebook NBC
News)



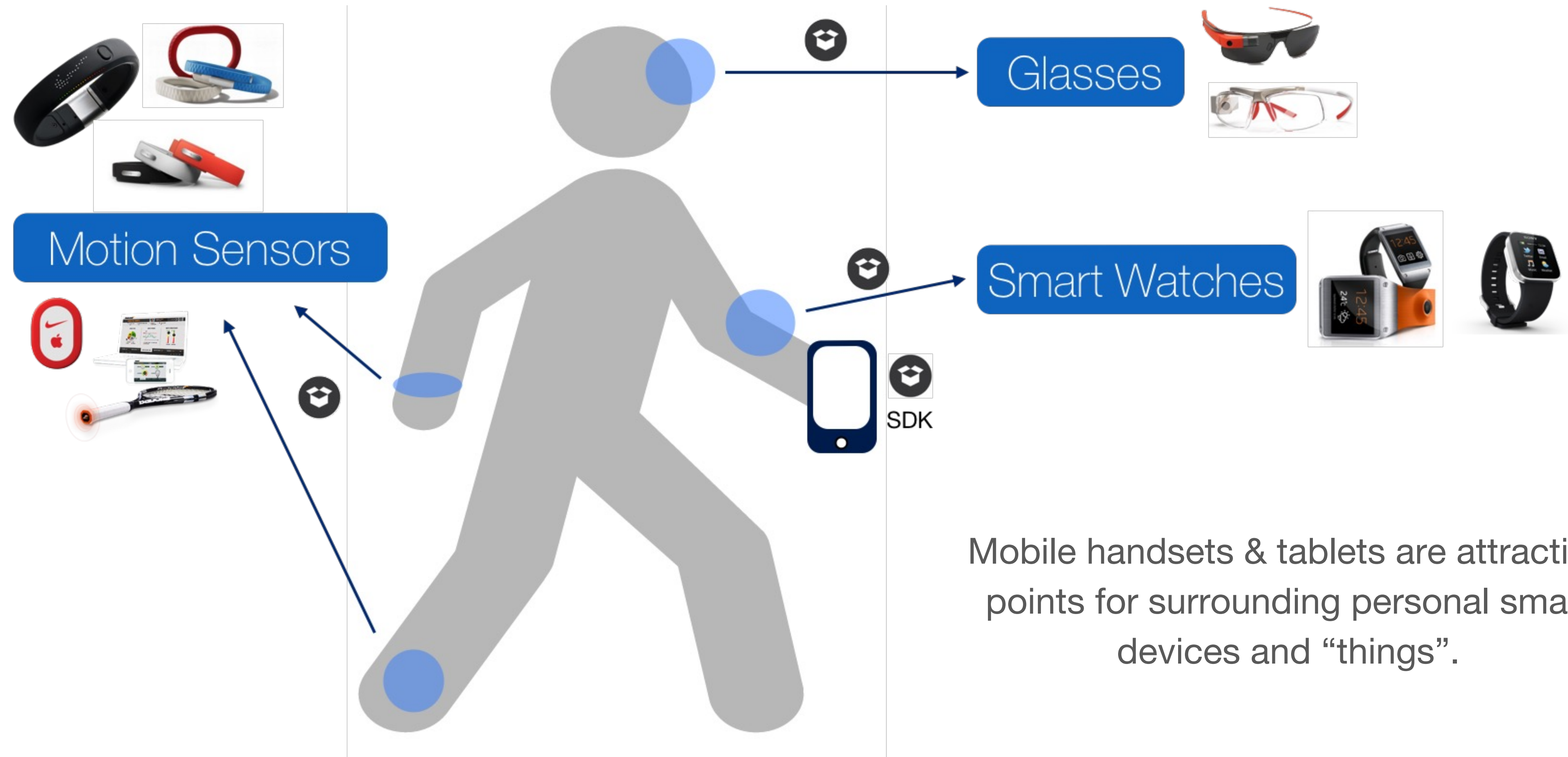
The Mobile Centric World



230K Devices at Vasco Rossi Concert @ Modena Park - July 2017

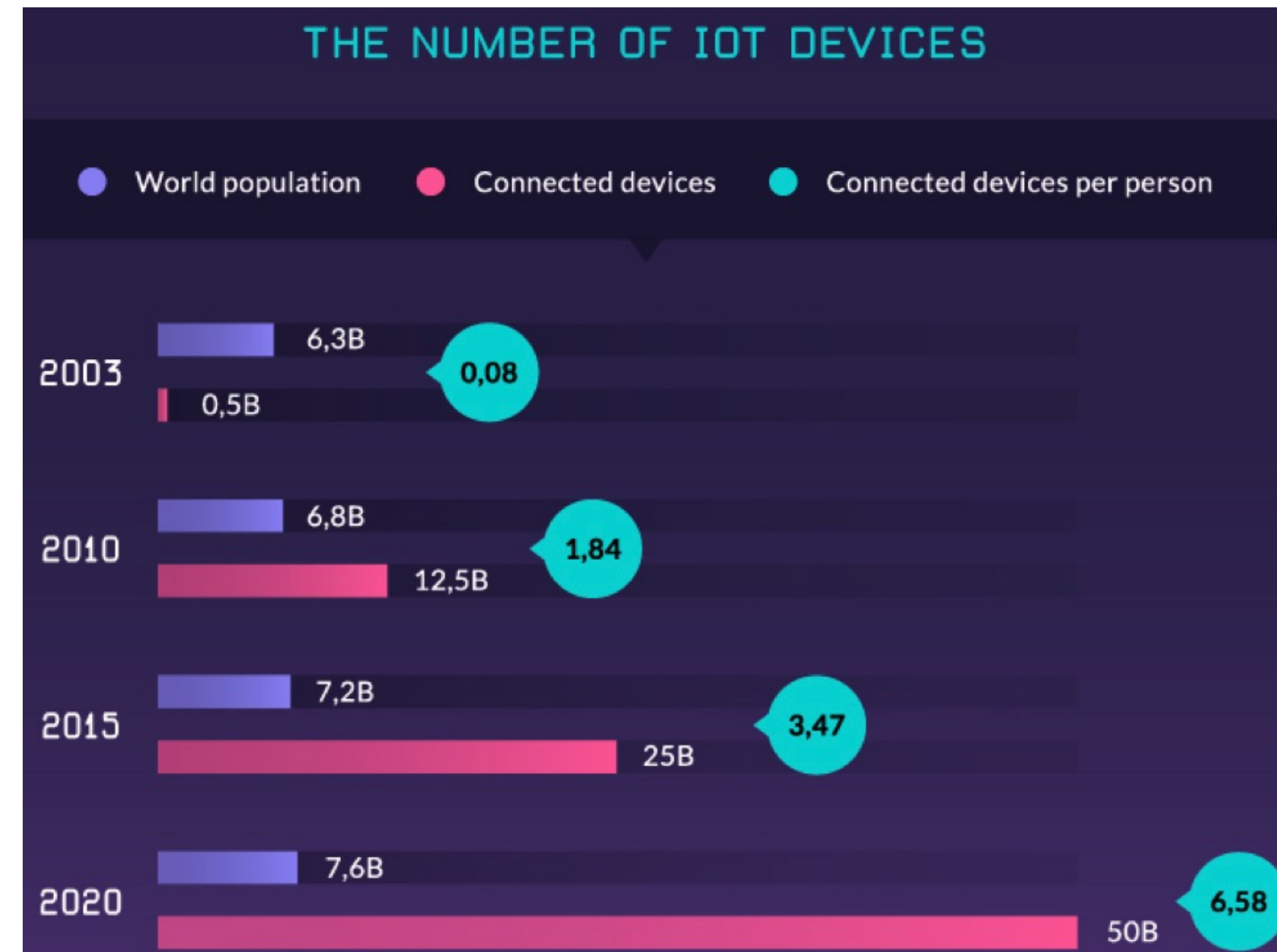
Source: <https://www.digitaldata.olivetti.com/timbigdata/concerto-di-vasco-rossi>

The Mobile Centric World



Mobile handsets & tablets are attraction points for surrounding personal smart devices and “things”.

Internet of Things: The Next “Big” Thing

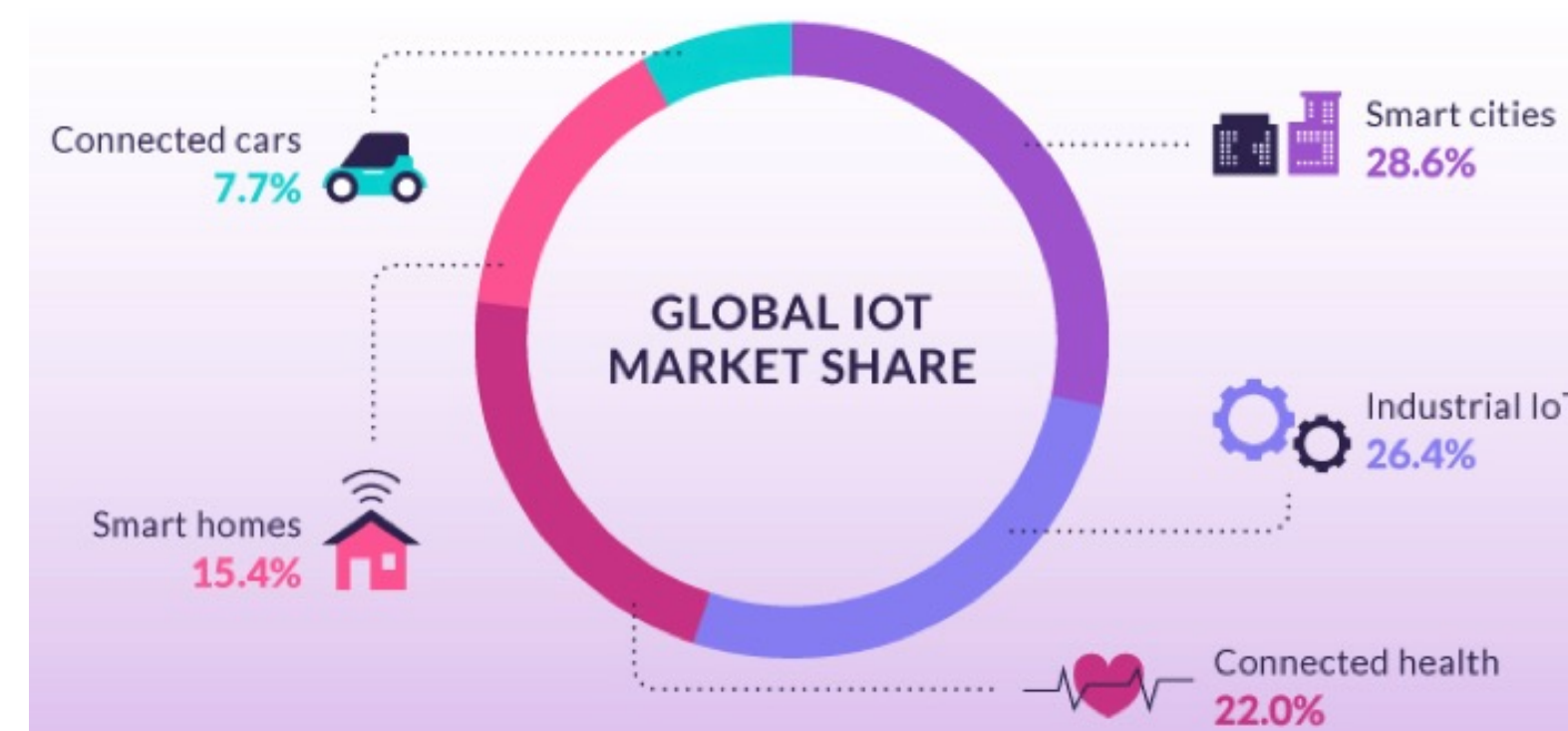


Internet of Things has already both *directly* or *indirectly* transformed several markets and fields such as:

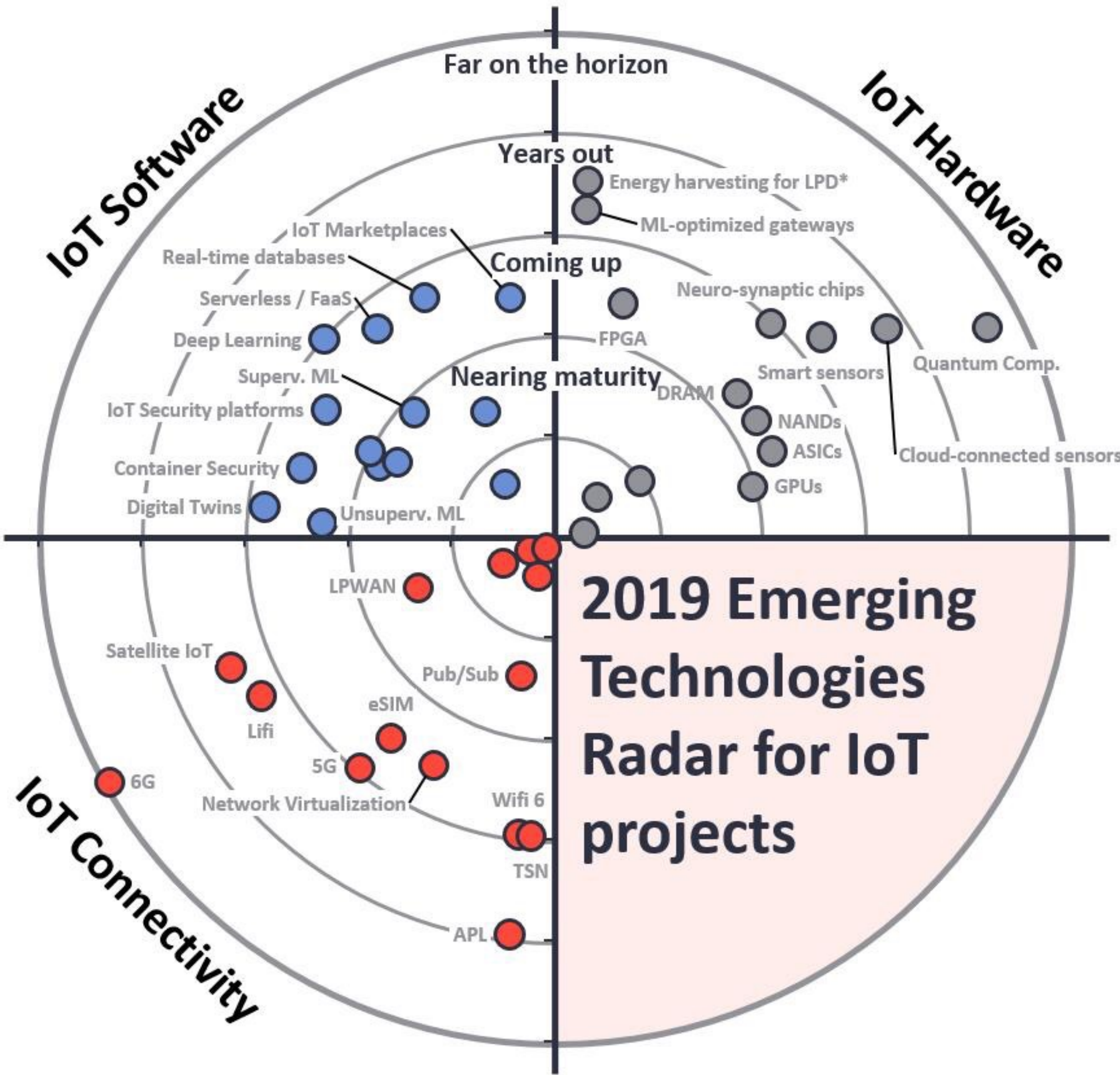
- Communication (e.g., 5G Cellular Networks)
- Internet Architecture (Edge Computing)
- Human–Machine Interaction
- Mobile Device and Applications
- Industry
- Single Board Computers and prototyping platforms

Furthermore it may embrace a massive amount of heterogeneous research and technologies fields at the same time:

- Computer Science
- Software Engineering
- Networking & Communication Systems
- Distributed Systems
- Artificial Intelligence and Machine Learning
- Electronics
- Business
- and counting



<https://safeatlast.co/blog/iot-statistics/>



Disclaimer: The 2019 Emerging technologies radar for IoT projects represents the analyst view of the IoT Analytics' research team under the disclaimer that depending on context the classification and perception of the project may be different in reality. IoT Software refers to Middleware, Analytics, Storage, Platforms, and Applications. LPD = Low power devices. The only relevant positioning of each topic is its distance to the center. The positioning on the axes plays no role.

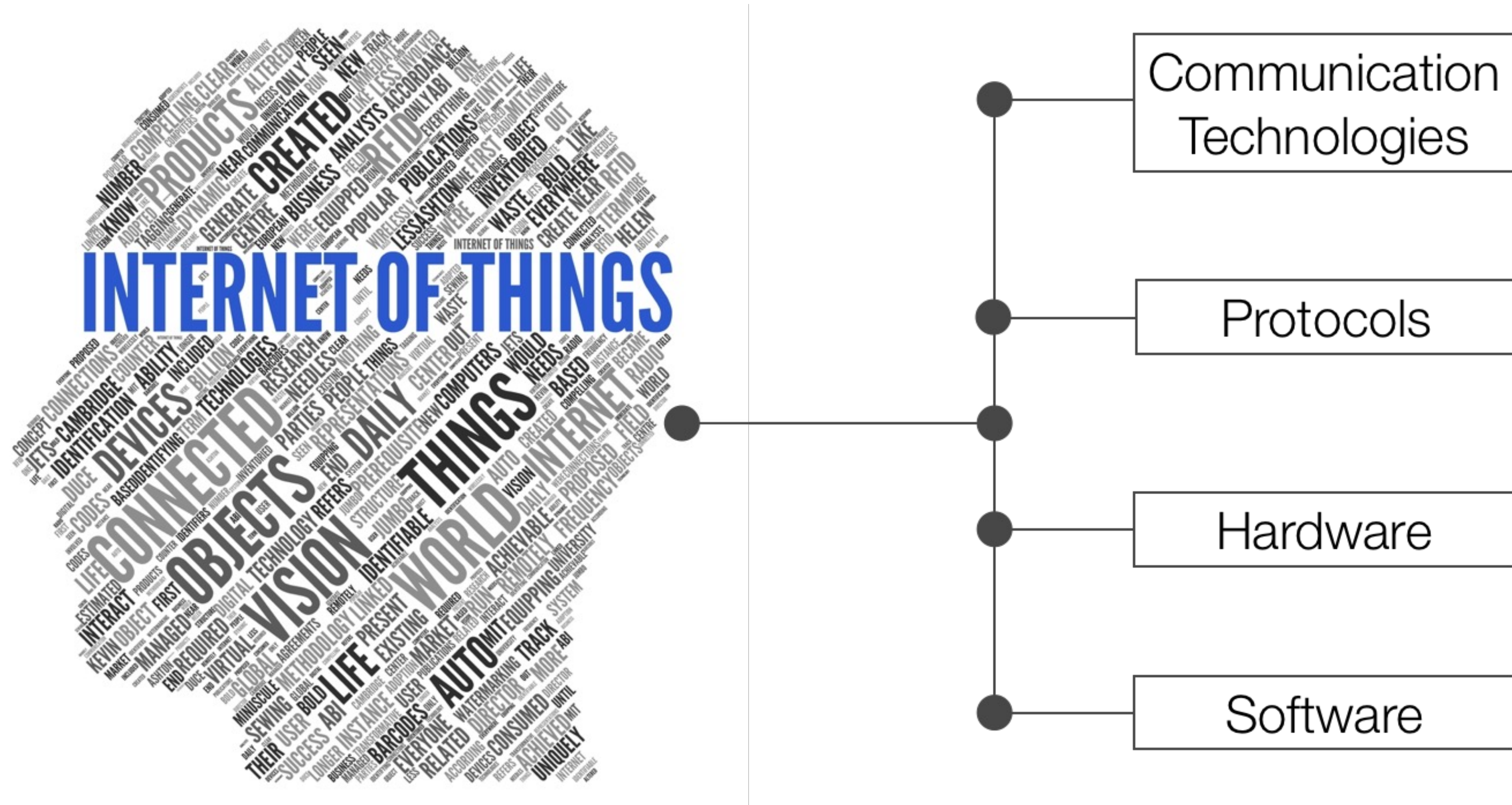
Source: IoT Analytics Research – August 2019



Smart Objects are items such as sensors, consumer devices, and enterprise assets that are connected to both the Internet and each other

The main and distinctive characteristics of the Internet of Things are the following

- Unique identifiable objects (things) with embedded intelligence able to sense or interact with the environment and communicate with the external world
- Presence of constrained devices with limited processing, storage and energy capabilities
- IP (Internet Protocol) based Networks (both IPv6 and IPv4)
- Different Communication Technologies (e.g., WiFi, Cellular, 802.15.4 etc ...)
- Standard protocols and Interoperability
- Disruptive heterogeneity in terms of data, hardware and software
- Security
- Interaction between People & Things



Connectivity & Connection

Heterogeneity Management

Interoperability & Interaction

Security

Data Collection & Processing

Standard Protocols Adoption

Real-time & Performance



Wireless Ad-hoc Sensor Networks (WSN) represent the ancestors of the Internet of Things without a common and shared IP network.

WSNs promised wide support of the interaction between people and their surroundings. The potential of the WSN lies in the three words composing the acronym itself:

- **Wireless:** puts the focus on the freedom that the elimination of wires gives in terms of mobility support and ease of system deployment
- **Sensor:** reflects the capability of sensing technology to provide the means to perceive and interact – in a wide sense – with the world
- **Networks:** gives emphasis to the possibility of building systems whose functional capabilities are given by a plurality of communicating devices, possibly distributed over large areas.

- Telemetry allows the possibility to collect remote measurements from the field
- Telemetry can be associated to:
 - **Long-range Measurements:** temperature, humidity, wind from remote weather stations (cellular network or Inmarsat satellite network); Formula 1
 - **Short-range Measurements:** pacemaker (telemetry coil, to measure the activity without surgery: low-power communication)
- *M2M is a generalization of telemetry, which implies autonomic bidirectional communication between non-human operated machines and a central collector device*



- Fragmented industry verticals:
 - most industries are solving their M2M needs on their own;
 - Often converging upon similar architectures and elements;
 - **the M2M “wheel” is reinvented from industry to industry**
- Customized solutions abound:
 - since solutions are on an industry-by-industry basis, **each solution requires its own design, production, and implementation cycle**;
 - unique solutions often result in sole-sourcing of **customized hardware**;
 - **software design is limited to a small group of developers** who understand the architecture and API's for a particular industry or platform, thus leading to high development costs and high costs for support.
- **Lack of scale for components:** economies of scale are difficult to achieve in any given industry since solutions are so varied and unique.

IoT vs Machine 2 Machine (M2M)

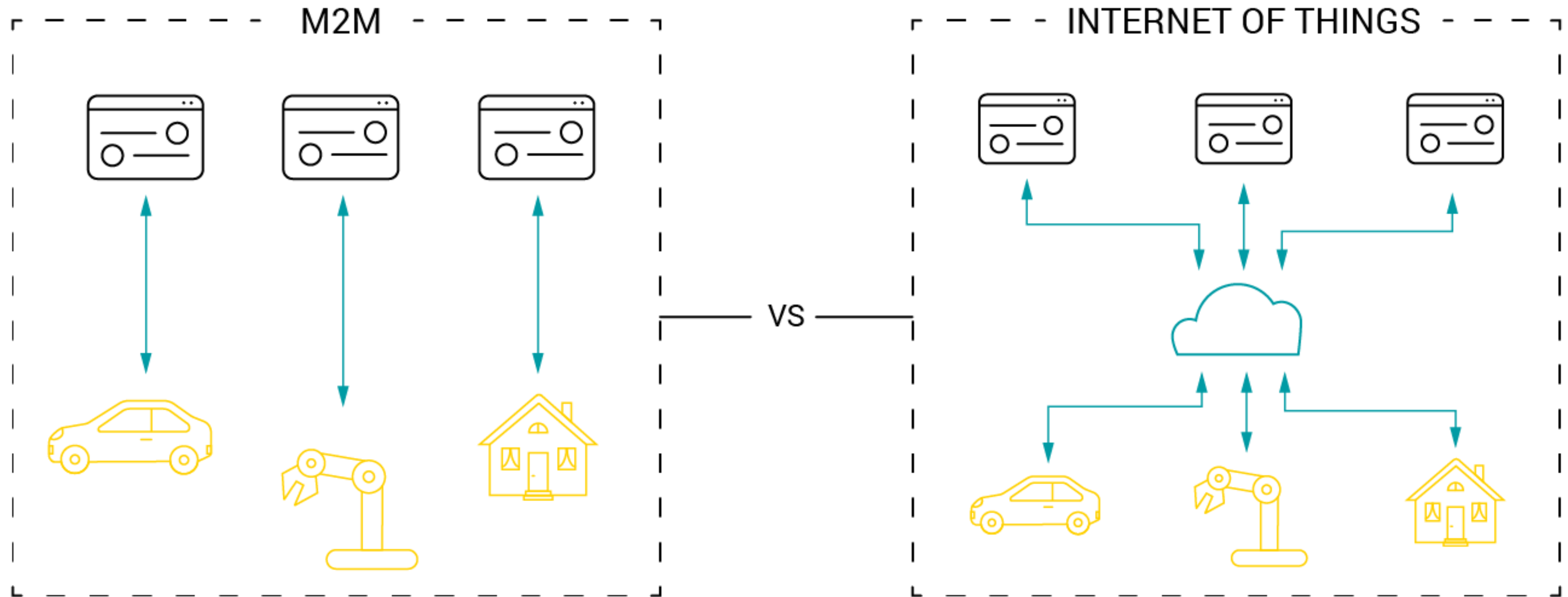
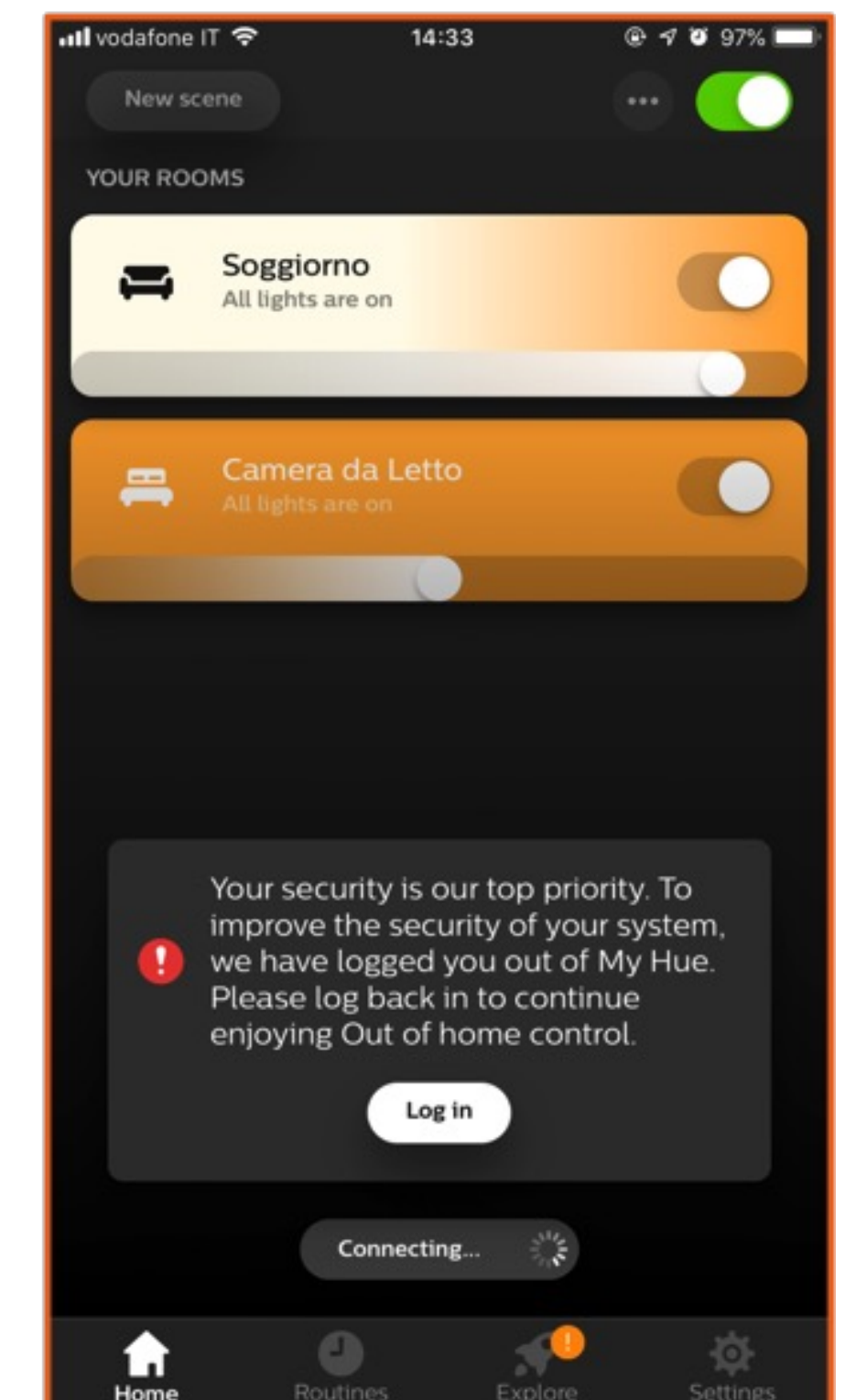
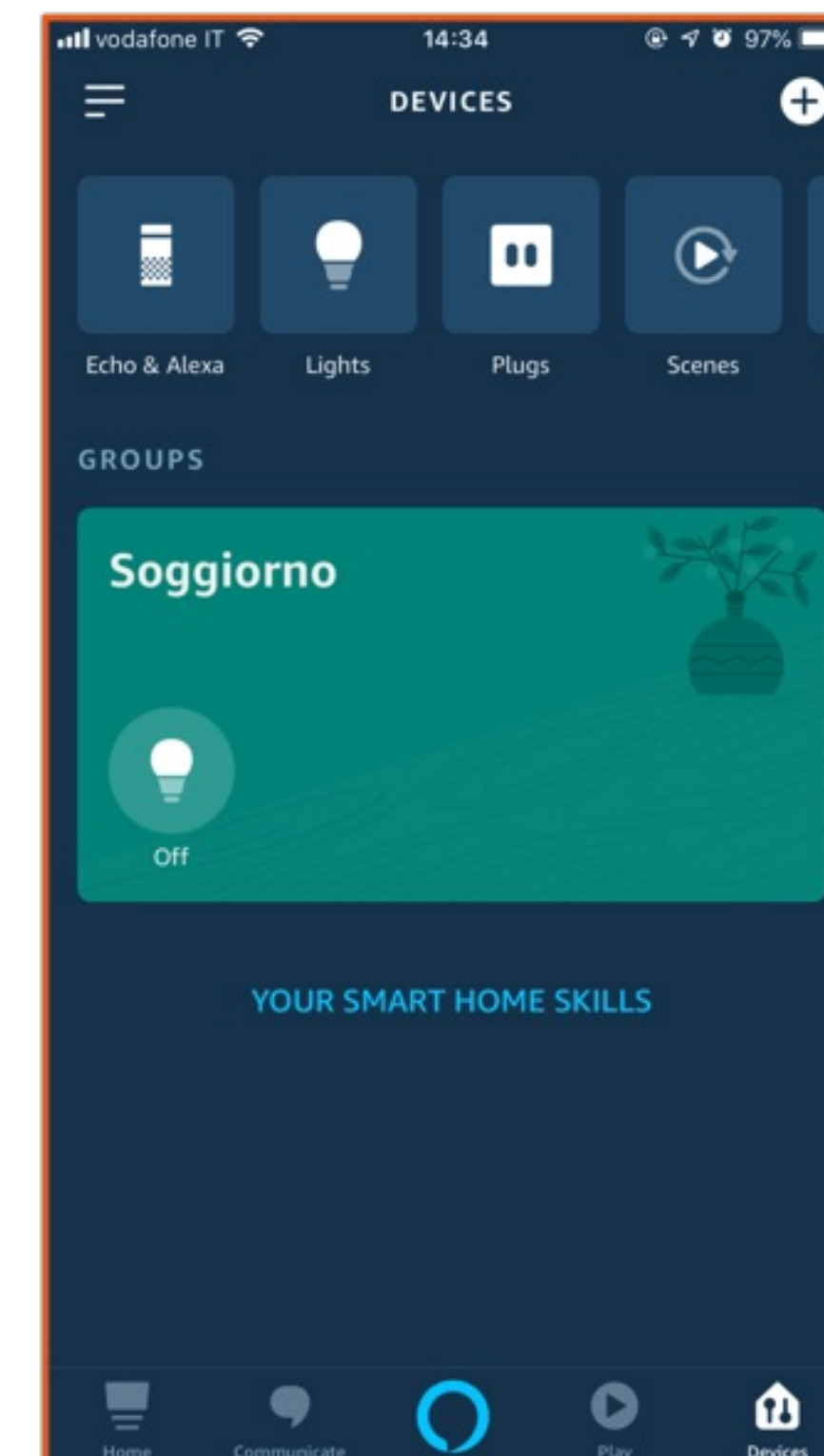
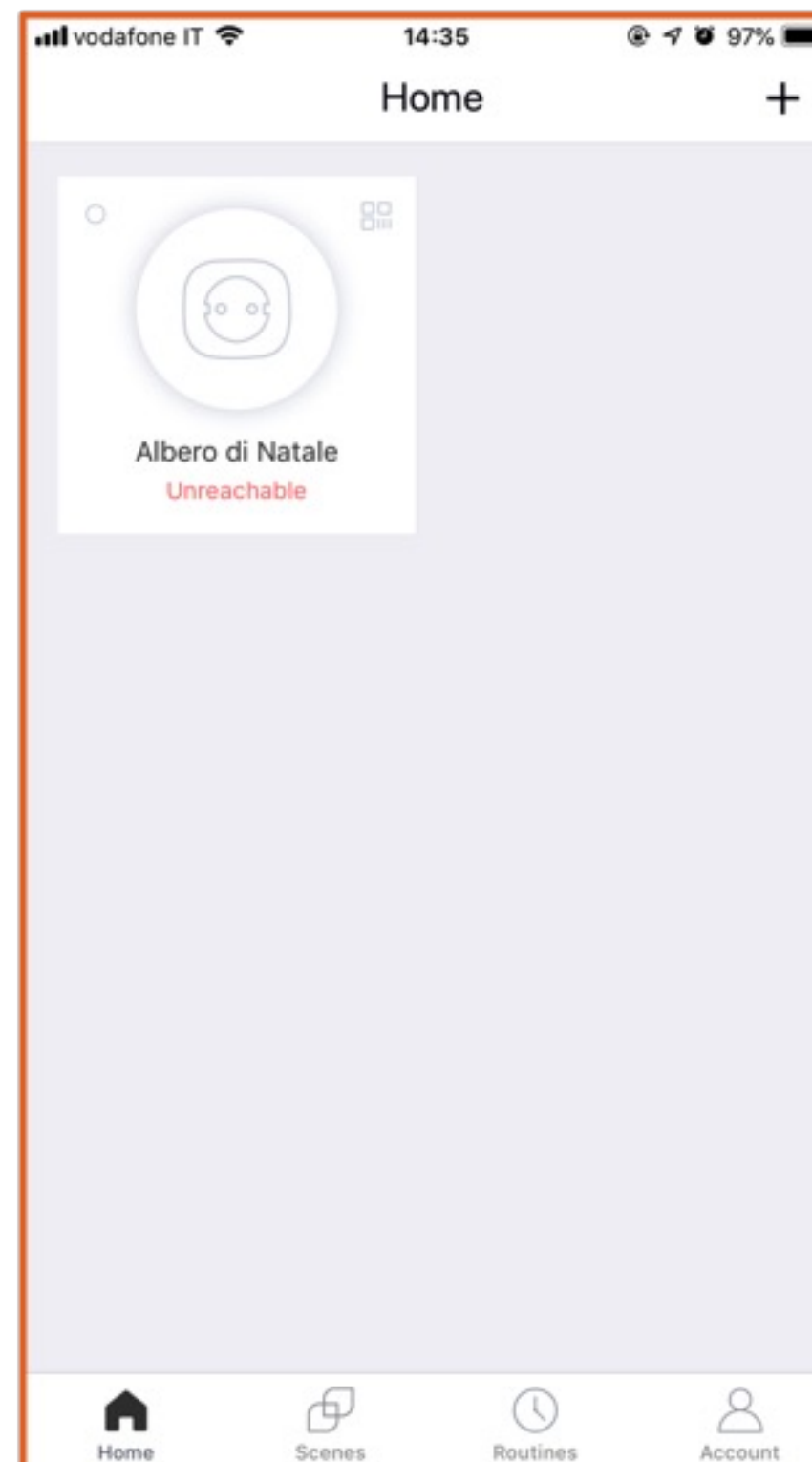
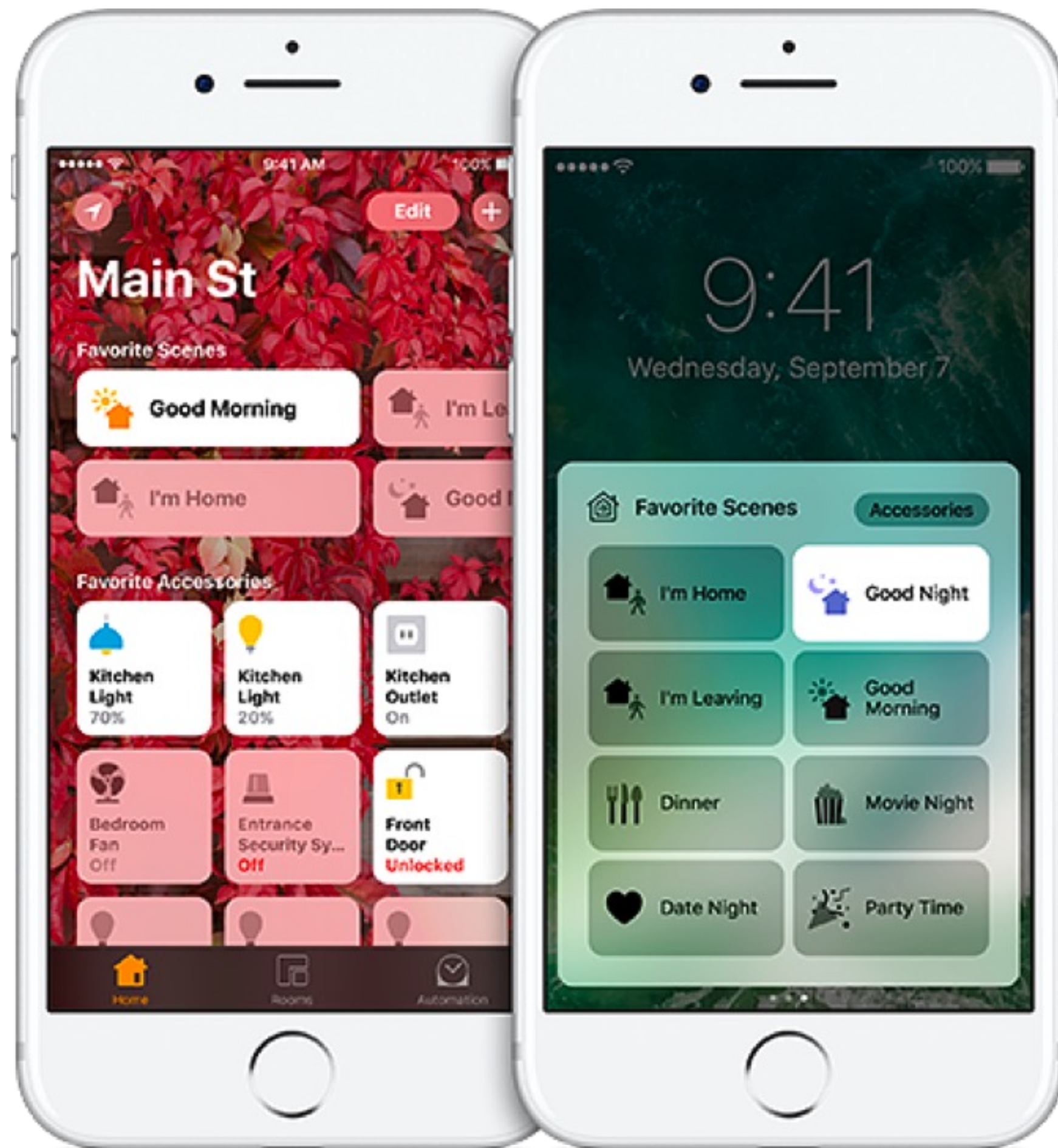


Figure from: <https://www.avsystem.com/blog/iot-and-m2m-what-is-the-difference/>

IoT vs Machine 2 Machine (M2M)

M2M	IoT
Simple device-to-device communication usually within an embedded software at client site	Grand-scale projects and want-it-all approach
Isolated systems of devices using same standards	Integrates devices, data and applications across varying standards
Limited scalability options	Inherently more scalable
Wired or cellular network used for connectivity	Usually devices require active Internet connection
Extensive background of historical applications	State-of-the-art approach with roots in M2M

Vertical Silos are not Real IoT !

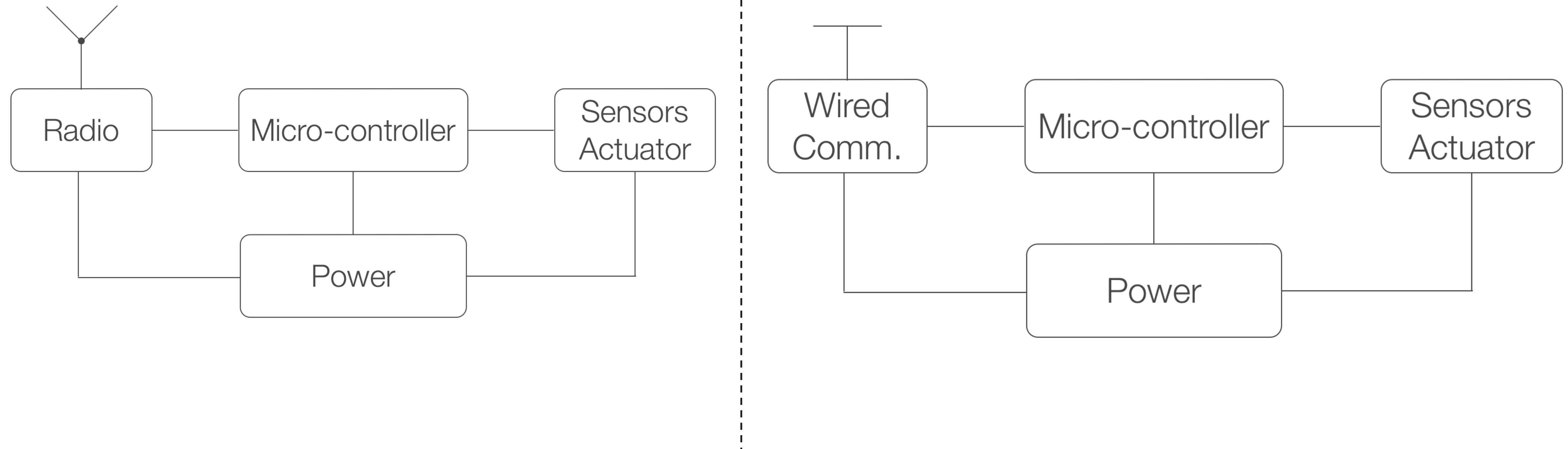


And counting

Vertical Silos are not Real IoT !



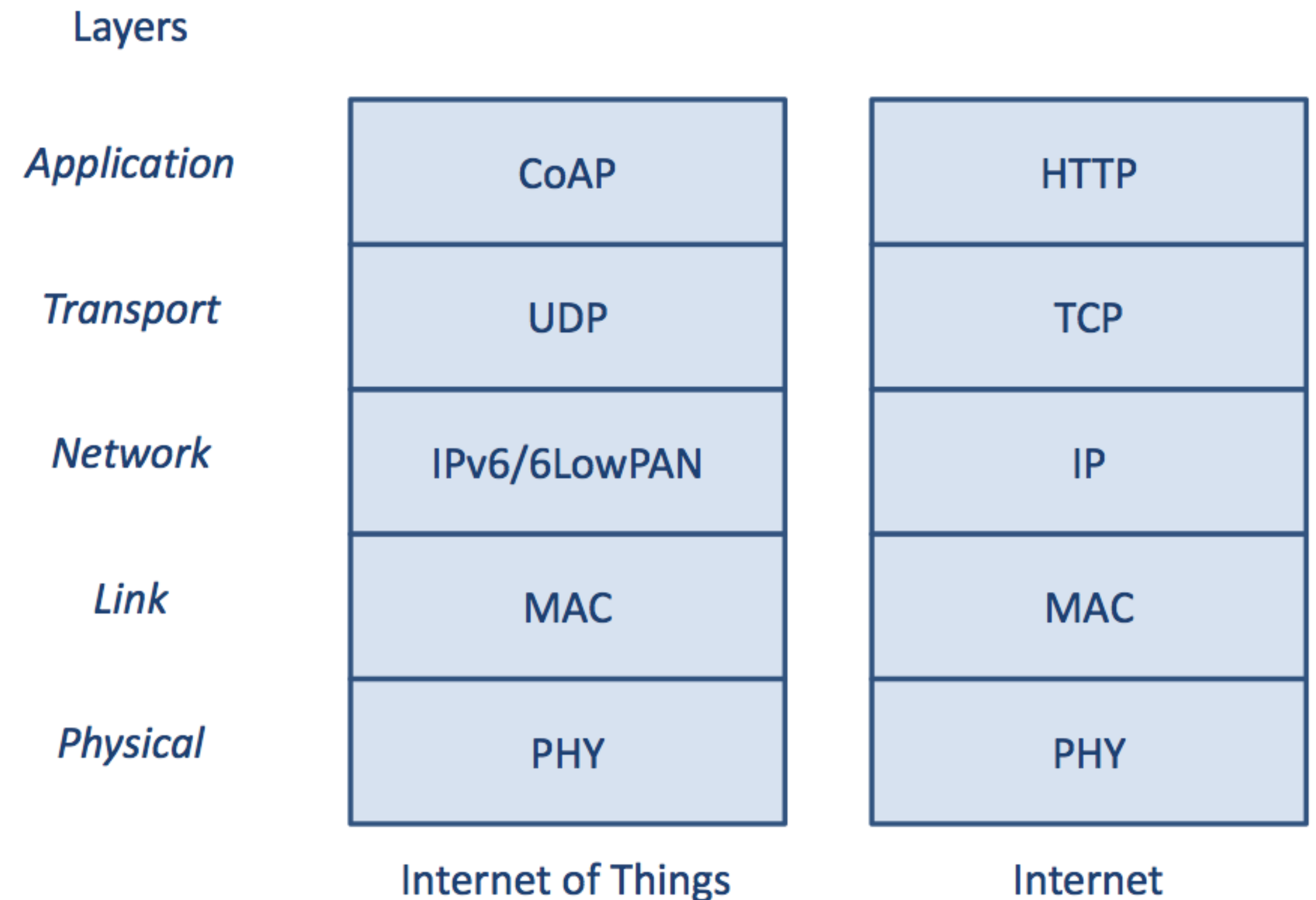
And counting

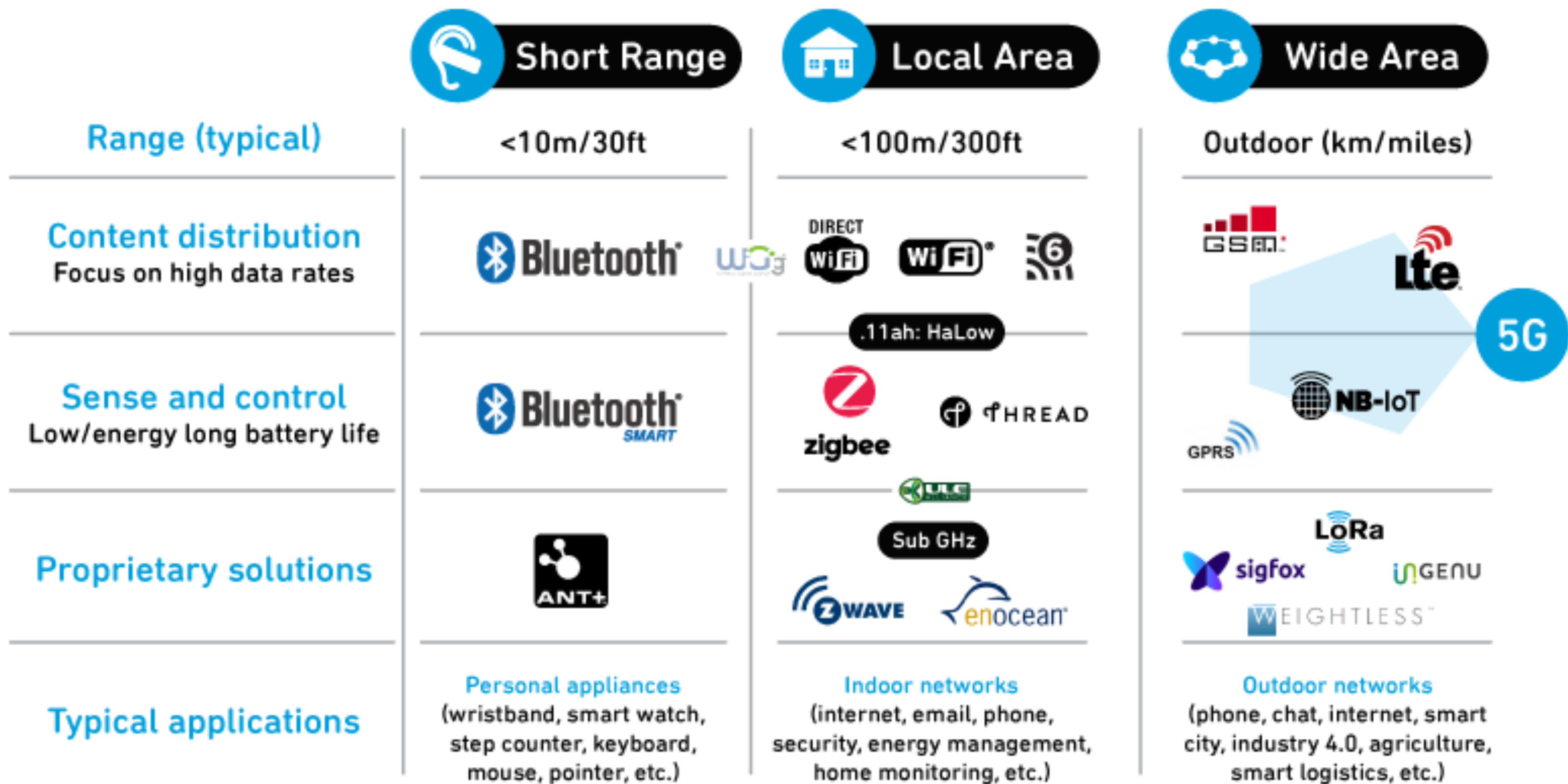


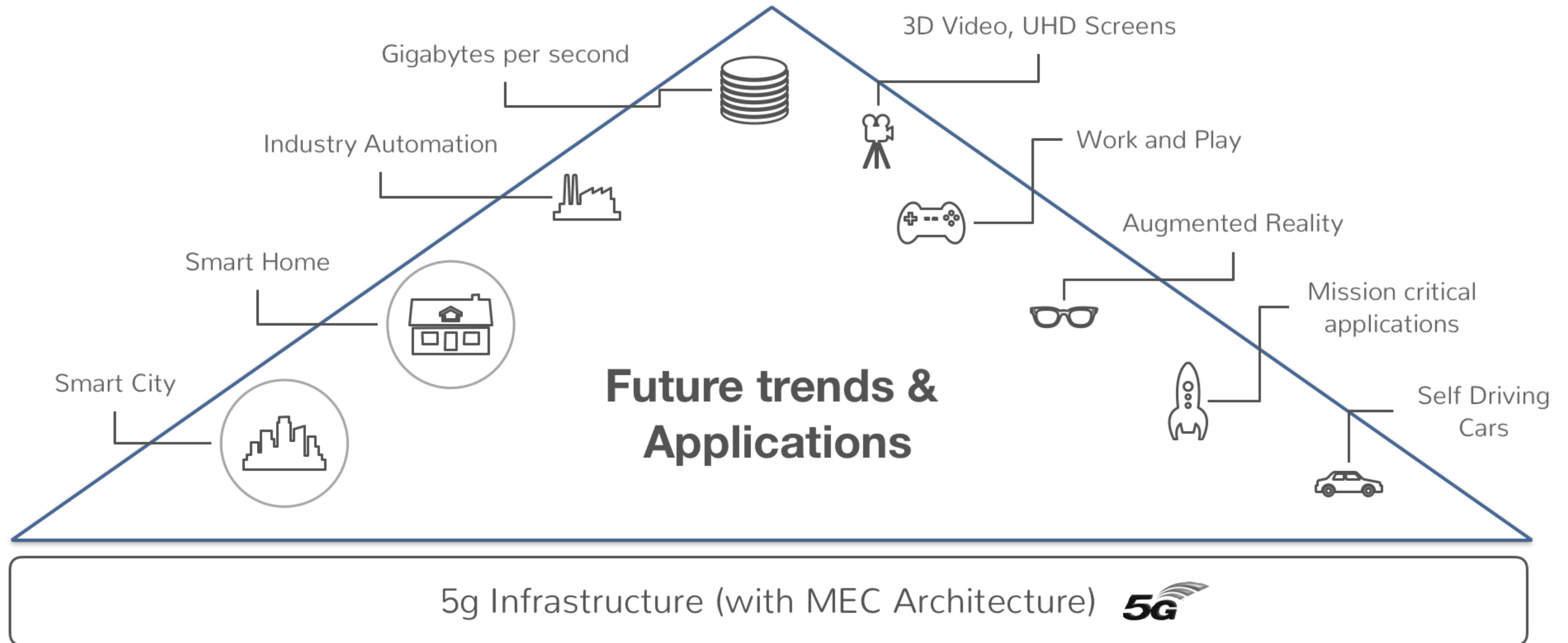
- **Communication:** This gives the smart object its communication capabilities. It is typically either a radio transceiver with an antenna or a wired connection.
- **Microcontroller:** This gives the smart object its behavior. It is a small microprocessor that runs the software of the smart object.
- **Sensors or actuators:** These give the smart object a way to sense and interact with the physical world.
- **Power source:** This is needed because the smart object contains electrical circuits. The most common power source is a battery, but there are other examples as well such as piezoelectric power sources, that provide power when a physical force is applied, or small solar cells, that provide power when light shines on them.

Internet of Things Protocol Stack

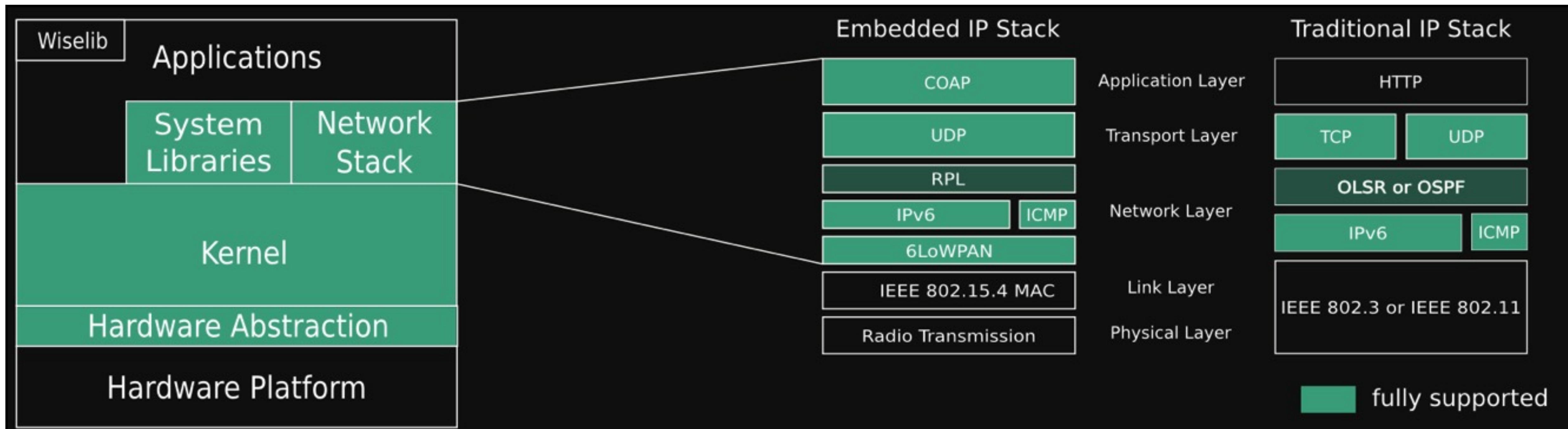
- The extremely high number of IoT devices, together with their heterogeneity, require proper mechanisms for the management and seamless interaction with smart objects.
- At the physical layer, low power Wi-Fi, **IEEE 802.15.4**, **LoRa** and **LoRaWAN**, **4.5G/5G**, and **Bluetooth Low-Energy (BLE)** are used to minimize energy consumption.
- Interoperability among IoT devices will be provided through the use of the **Internet Protocol (IP)**, which will also guarantee integration with the Internet (**6LoWPAN**).
- At the application layer, **CoAP (RFC 7252)** has been designed to bring the REST paradigm to constrained devices. **MQTT** represents instead the Pub/Sub IoT standard at the moment.





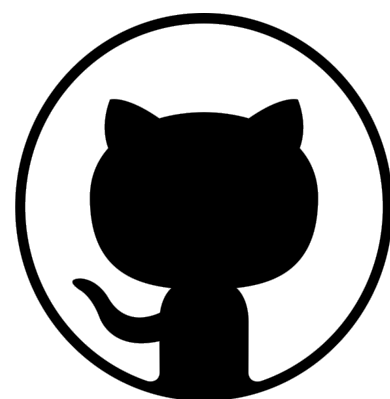
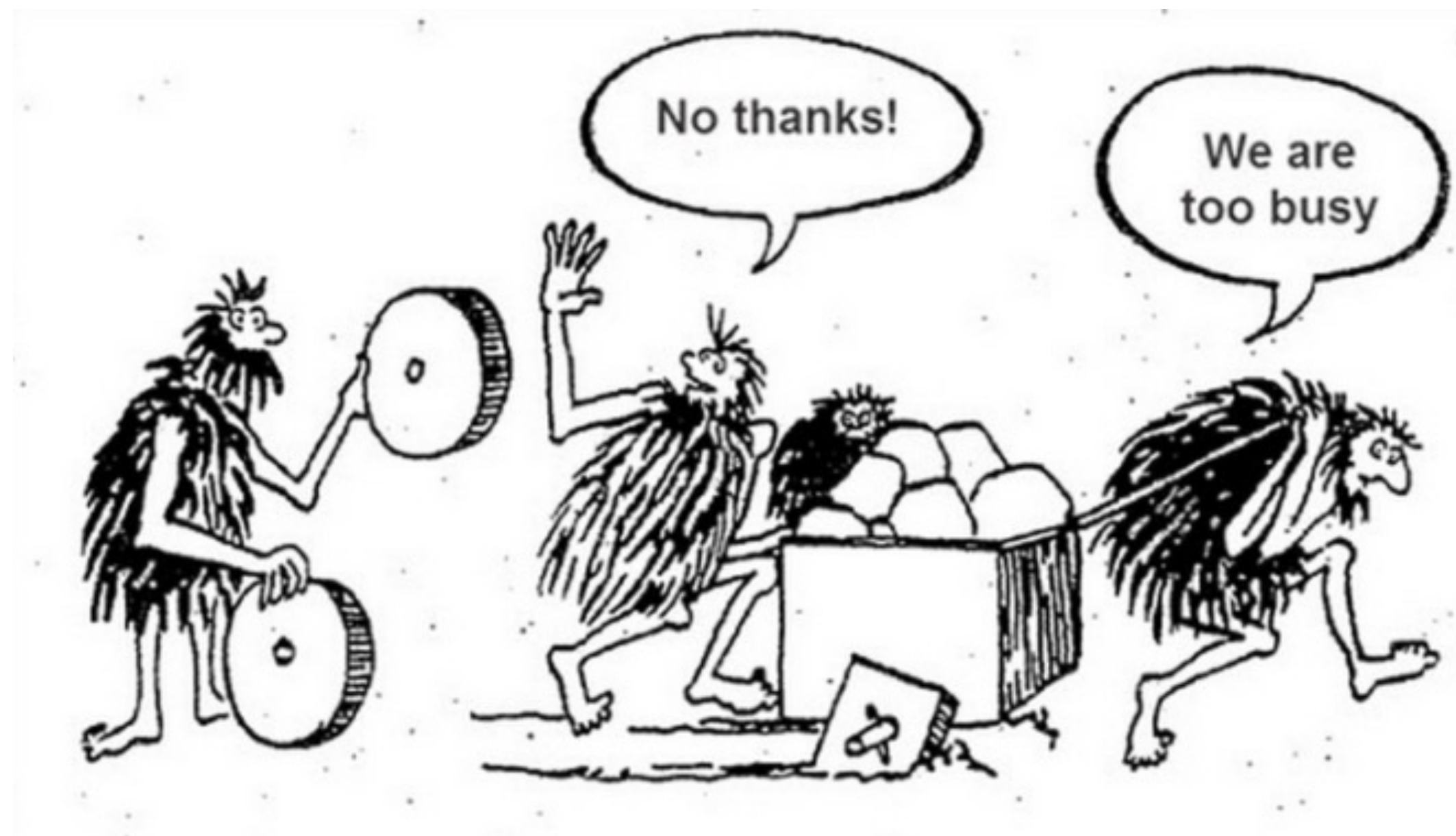


IoT Software - Just an Example



Source and Example from: <http://www.riot-os.org/>

IoT Software - Do not Reinvent the Wheel



CoAP

Repositories	1K
Code	?
Commits	48K
Issues	8K
Packages	0
Marketplace	0
Topics	32
Wikis	1K
Users	44

Languages	
Java	233
C	160
JavaScript	160
Python	115
C++	77
Go	44
HTML	33
C#	29
Makefile	26
Shell	17

MQTT

Repositories	22K
Code	?
Commits	188K
Issues	63K
Packages	4
Marketplace	0
Topics	207
Wikis	7K
Users	132

Languages	
JavaScript	3,827
Python	3,507
C++	3,139
Java	2,743
C	1,336
Go	770
C#	528
HTML	438
Shell	350
TypeScript	243

IoT

Repositories	67K
Code	?
Commits	376K
Issues	88K
Packages	1
Marketplace	0
Topics	445
Wikis	9K
Users	10K

Languages	
JavaScript	10,589
Python	7,892
C++	7,136
Java	6,132
C	3,344
C#	3,019
HTML	2,807
CSS	1,233
PHP	992
TypeScript	854

IoT Applications - One Point of View

Machine-to-Machine

Sensor Stream

Human-to-Machine

Crowdsourcing

Information-to-Machine

Internet

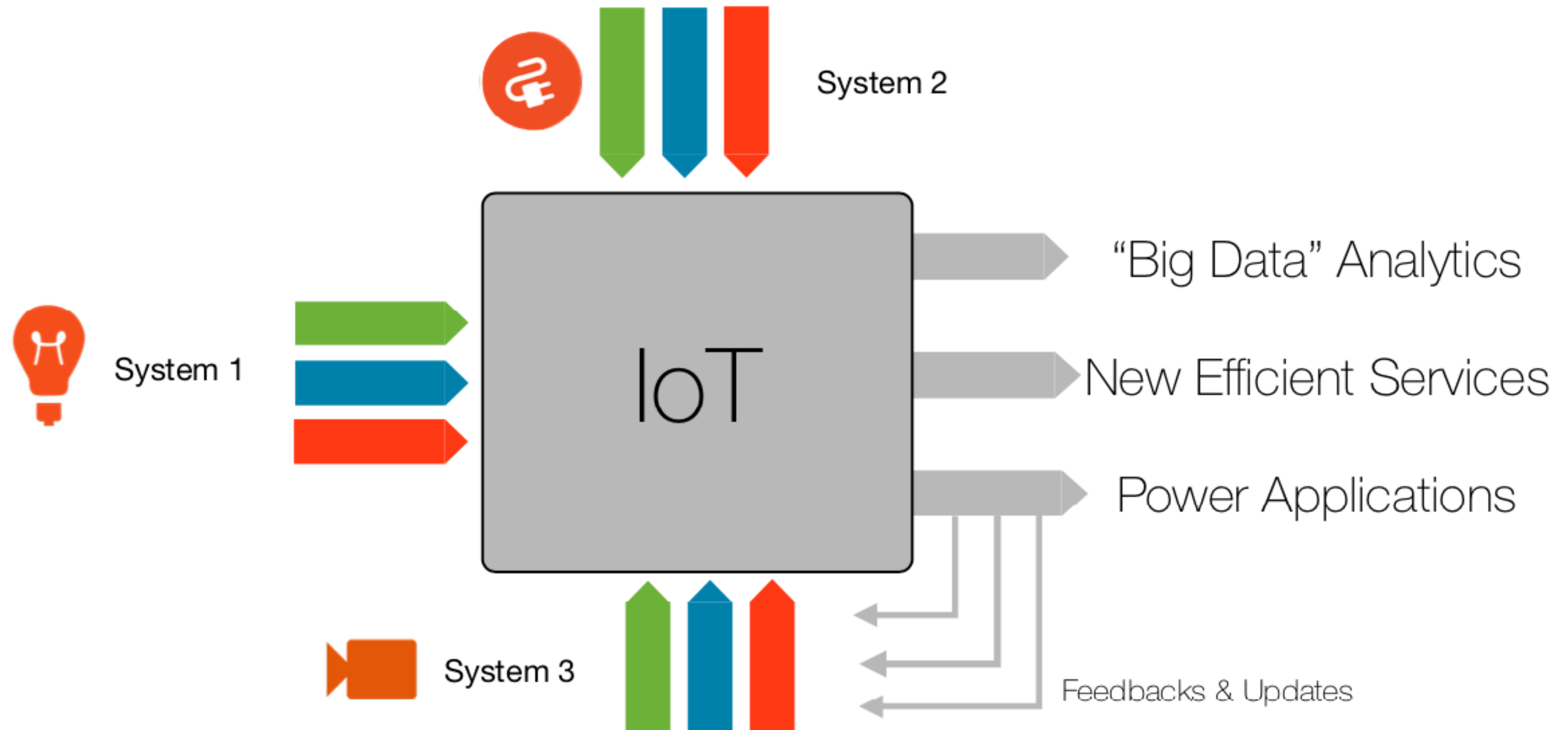
Cloud Computing
&
“Big Data” Analytics

Improve Efficiency

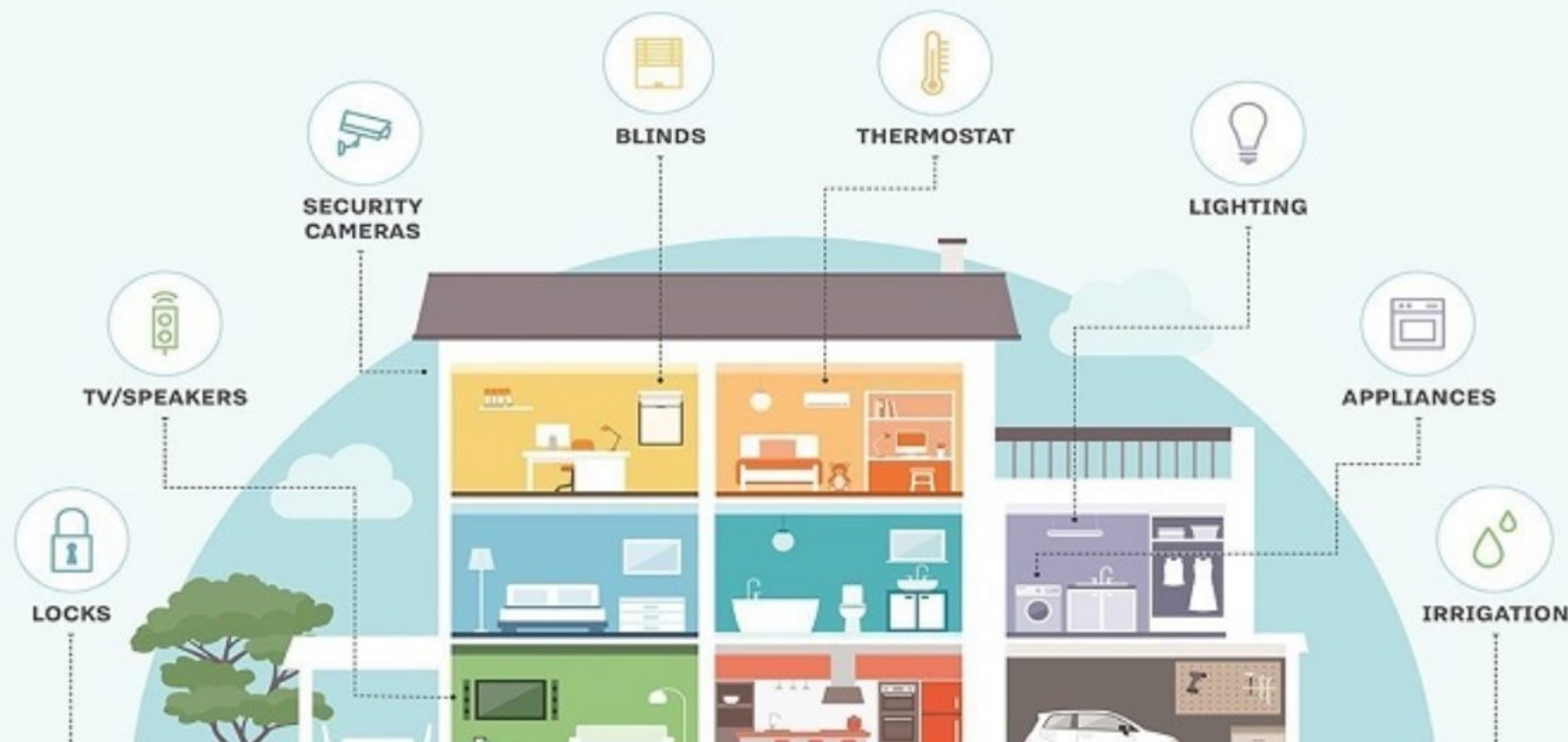
Offer New Services

Power Application

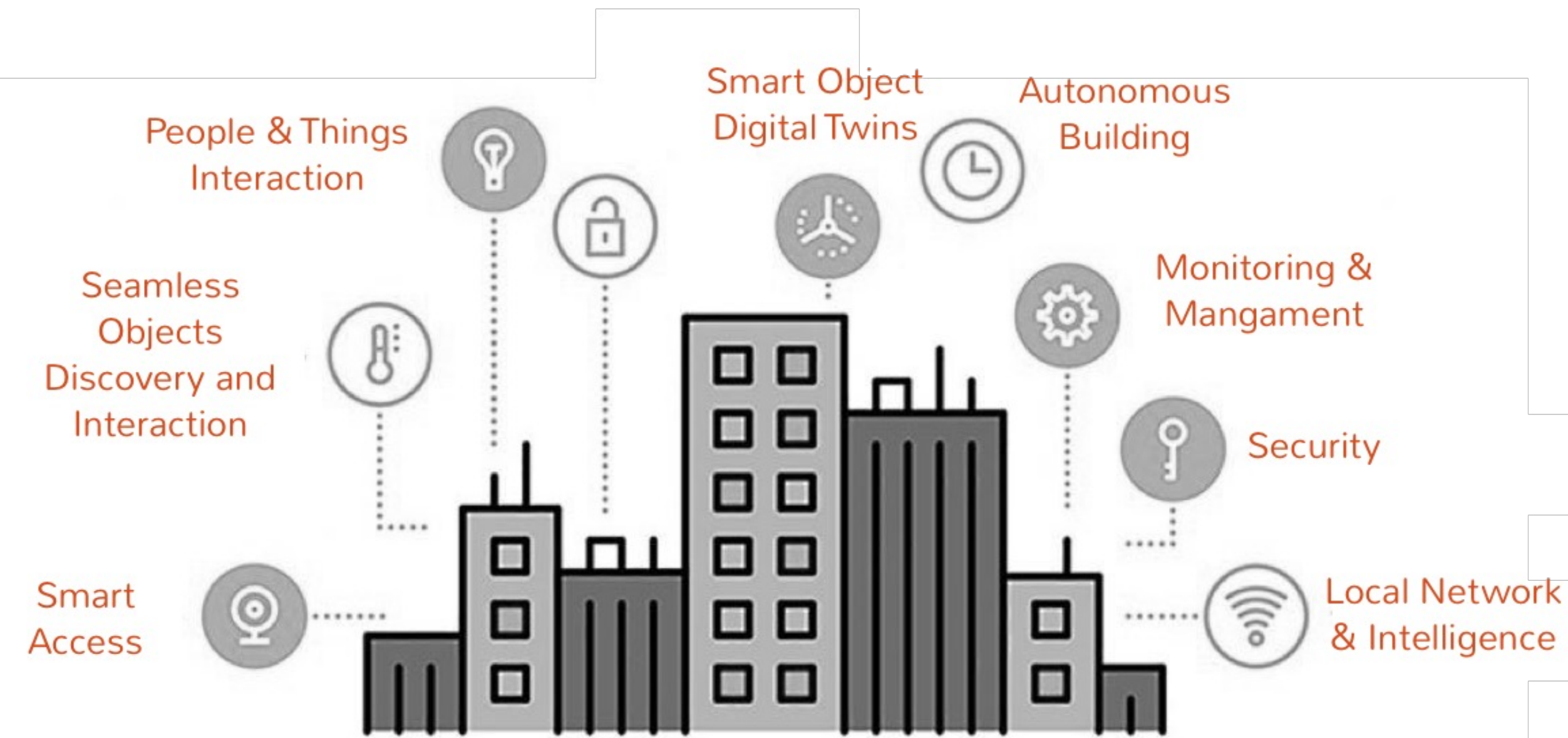




Smart Home & Smart Building



- Apartment Local Control
- Local Dynamic Policies
- Limited number of guests
- Simple Behaviour and limited number of Smart Objects



- Centralized Building Control
- Common Policies (e.g., Office hours, Temperatures, etc ..)
- High number of guest, visitors and promiscuous accesses
- Complex Behaviour and High number of heterogeneous devices and data

Services

- Physical and IT Security (for each architectural layer)
- **Granular Authentication and Authorization**
- **Smart Object Access Delegation**
- Telemetry & Monitoring
- Energy and System Efficiency
- **Human - Machine Interaction**
- **Presence Detection**

Connectivity

- Short Range (Bluetooth)
- Local Area (Ethernet, WiFi, ZigBee/802.15.4)
- Local and Remote High Bandwidth (Mbits/Sec)
- Wide Area Connectivity (LoRaWAN, 4.5G)

Constraints

- Heterogenous Objects, Services and Protocols
- Combination of Standard and Proprietary products and services
- Coordination and Integration of existing and new appliances
- Local Intelligence and “Offline” Capabilities

Users & Targets

- Building Owner & Staff
- House Owner & Family
- Guests
- Visitors



- Heterogeneous Applications, Data and Connectivity Solutions
- Interoperability and Seamless Communication between subsystem is a key of success though the use of standard protocols, Open Data and a shared technological stack
- Mobility of users and things
- Citizens, Visitors and Tourists as natural consumers of customized data and services

Services

- Physical and IT Security (for each architectural layer)
- Granular Authentication and Authorization
- **Core Services and Infrastructure**
- **Mobility**
- Telemetry & Monitoring
- Energy and System Efficiency
- **Tracking and Presence (Payment, Security, etc ...)**

Connectivity

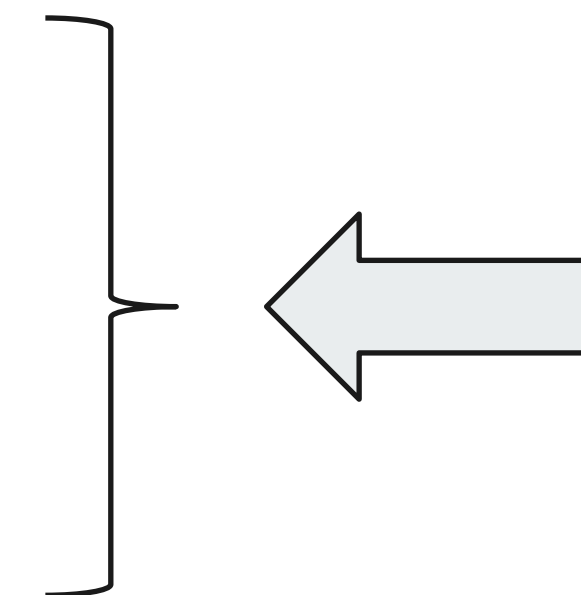
- ~~Short Range (Bluetooth)~~
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Constraints

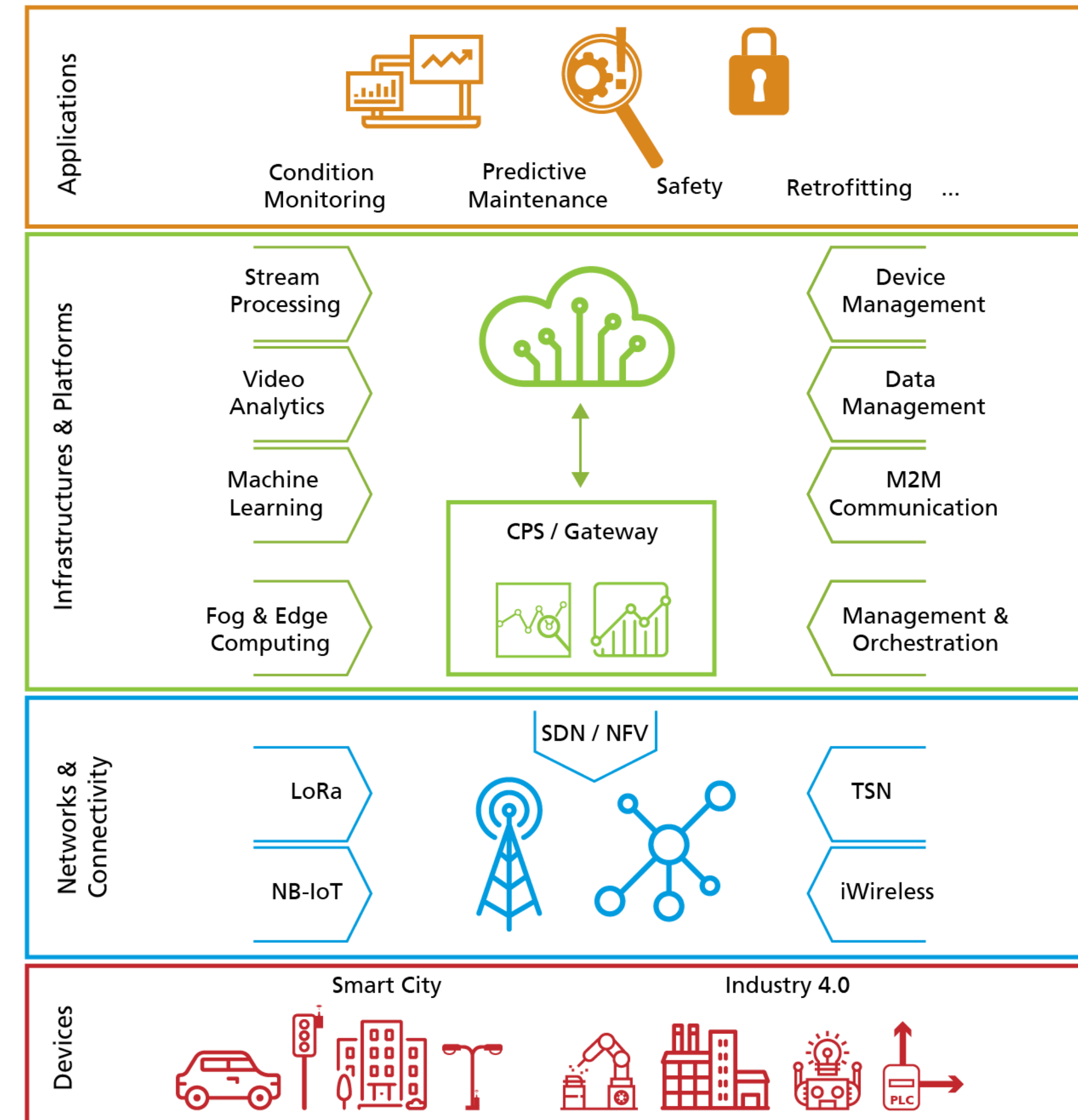
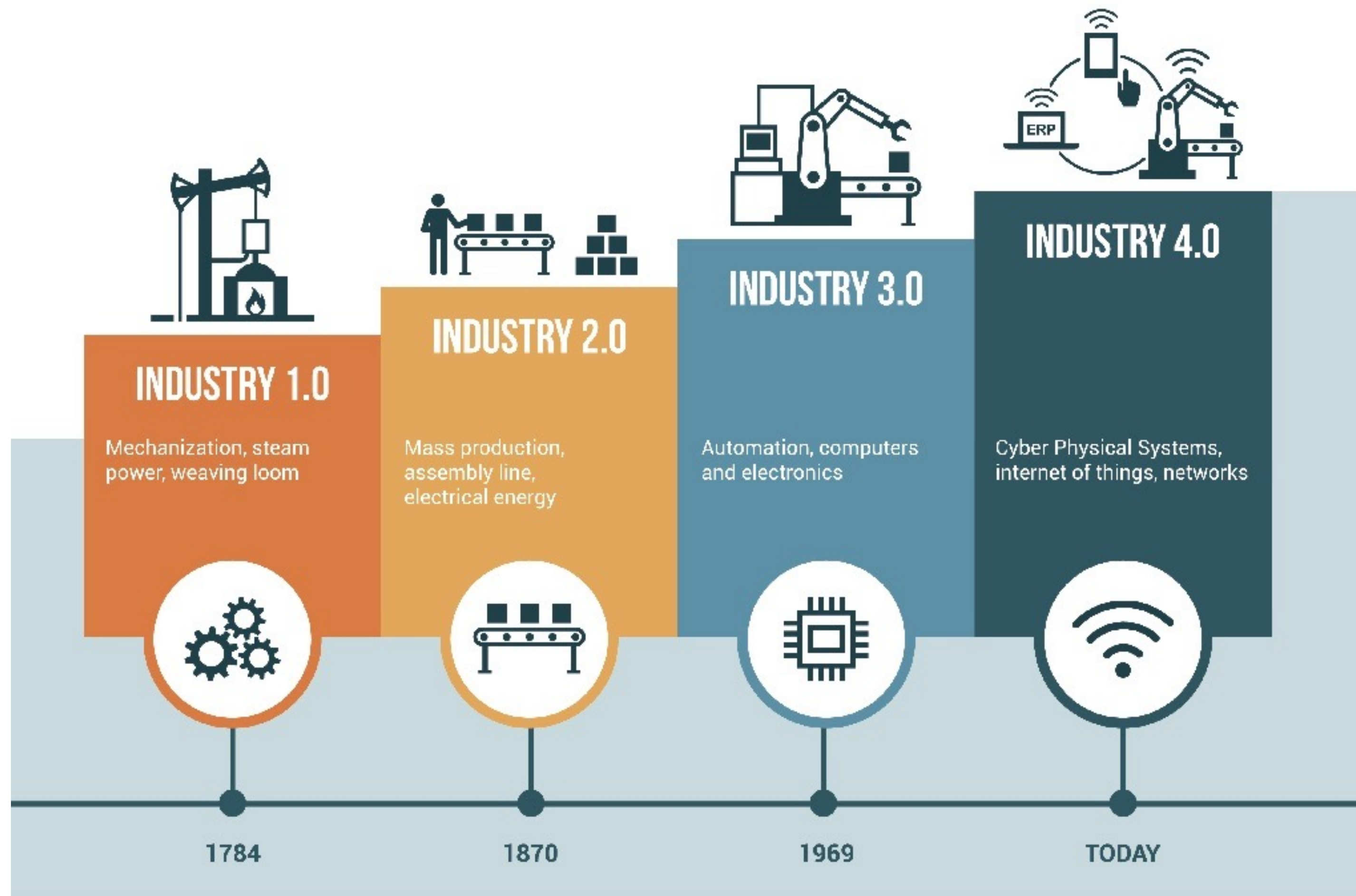
- Heterogenous Objects, Services and Protocols
- **Combination of multiple providers, services and datasets.**
- **The coordination is delegated to the Municipality**
- **“Local” Intelligence and “Offline” Capabilities**

Users & Targets

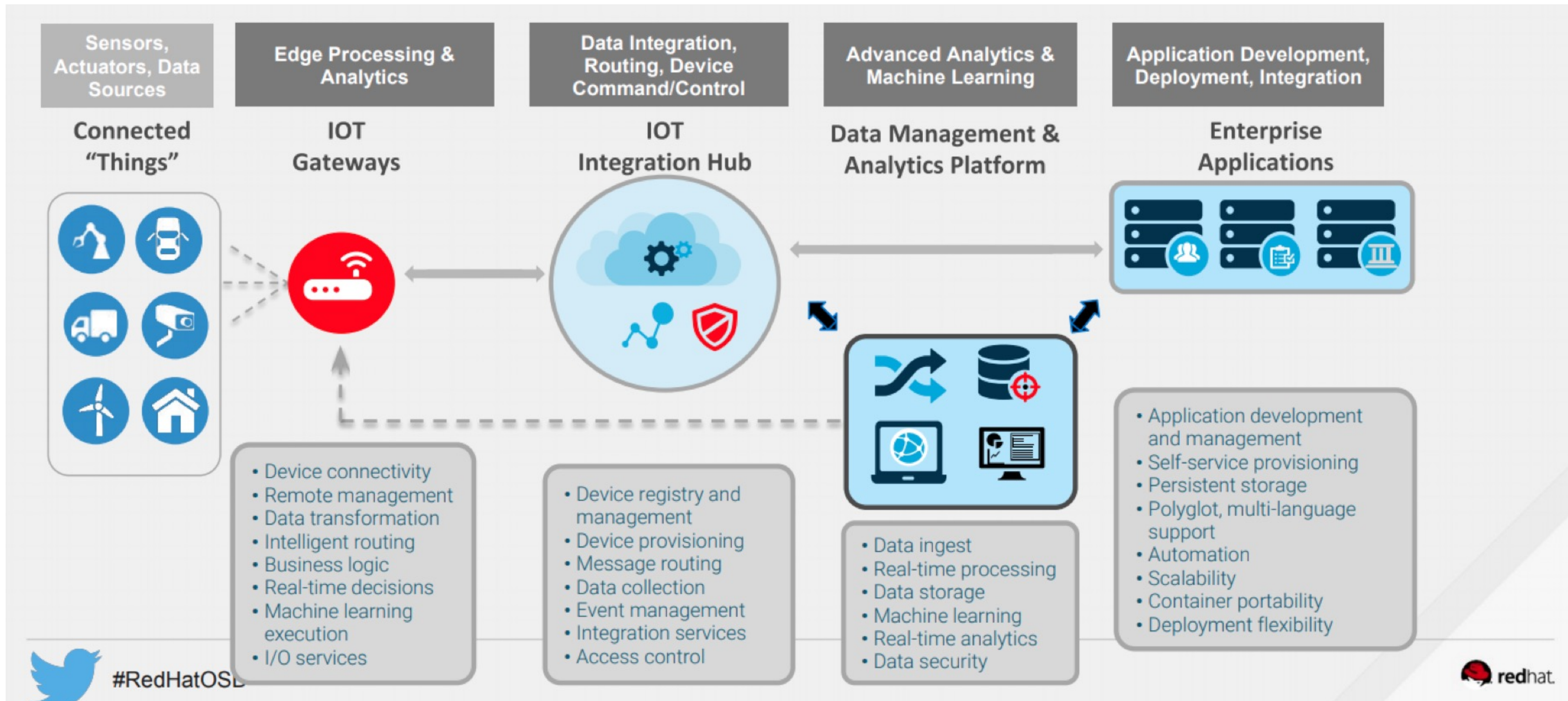
- Citizens
- Tourists
- Visitors
- Operators & Workers



Industrial IoT (IIoT)

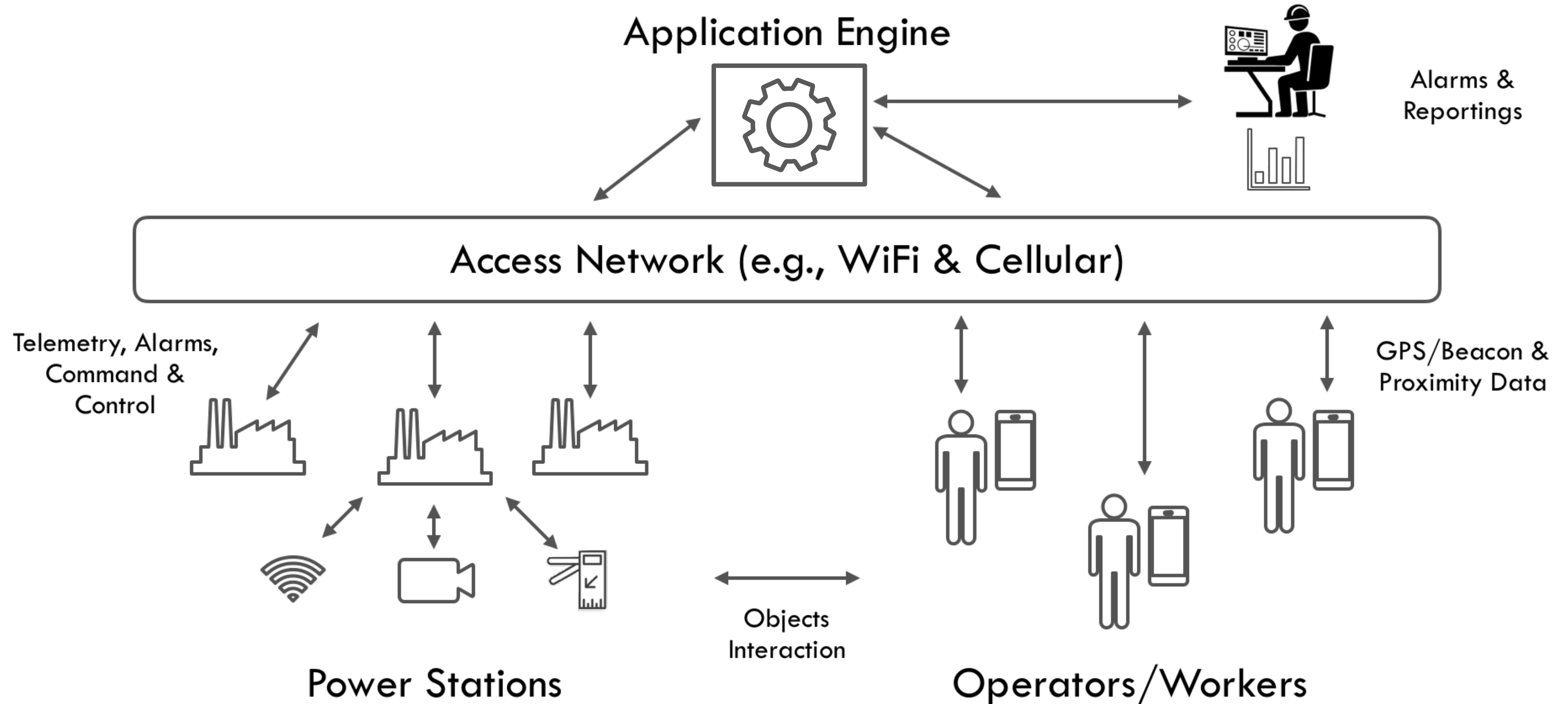


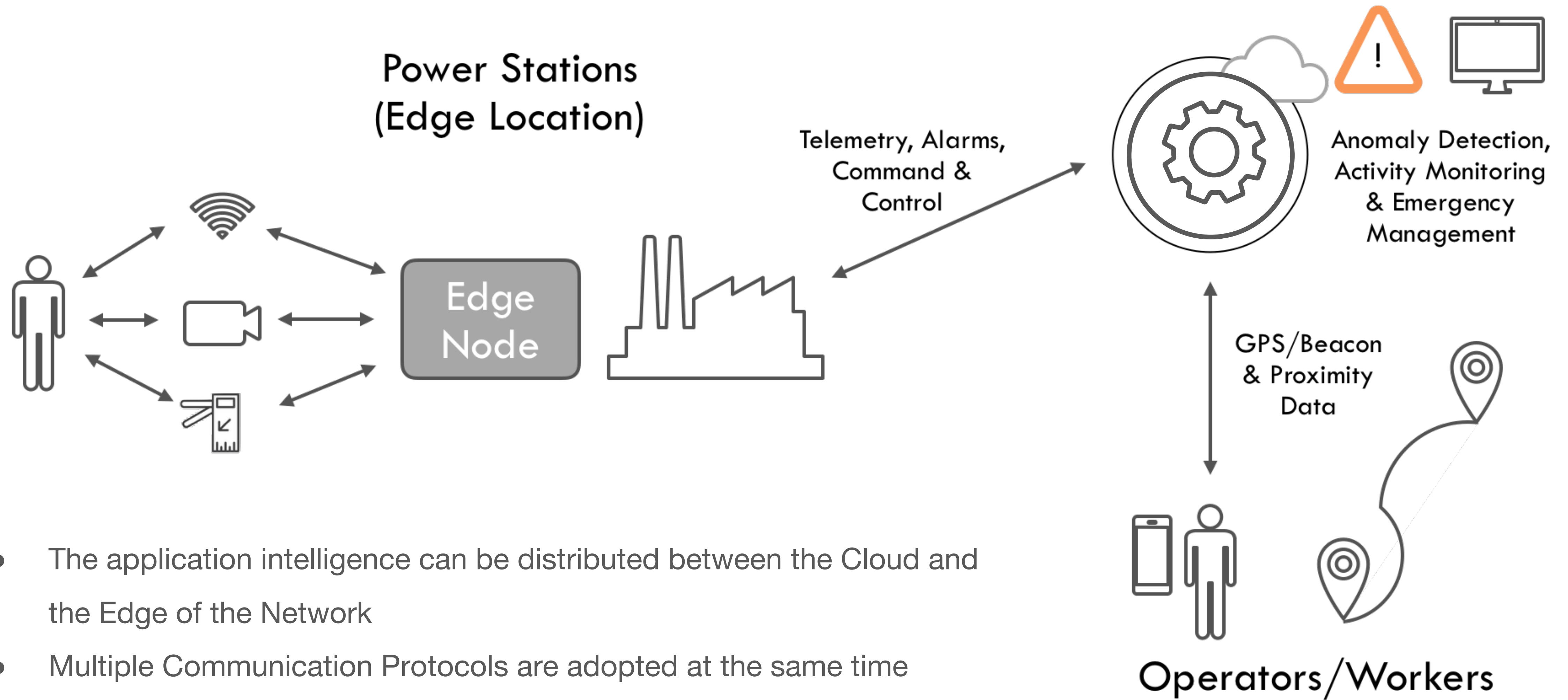
Industrial IoT (IIoT)



Source: RedHat l'OpenSource e i suoi partner per l'Internet of Things. Milano 30/10/2018, RedHat Forum 2018.

Fleet & Assets Monitoring





- The application intelligence can be distributed between the Cloud and the Edge of the Network
- Multiple Communication Protocols are adopted at the same time
- Local Interaction between People & Things on the Edge



In Fondazione UniverMantova una tre giorni dedicata allo sviluppo di idee e soluzioni innovative per affrontare le sfide legate alla gestione sostenibile della risorsa idrica nel territorio mantovano. L'iniziativa si inserisce all'interno del progetto "Acquam – Mantova, una provincia tra terra e acqua" e aprirà ufficialmente la Notte Europea della Ricerca 2026.

DESTINATARI

L'hackathon si rivolge a studenti universitari, neolaureati, informatici, sviluppatori, maker, architetti, designer, pianificatori, economisti, manager, esperti di comunicazione, ricercatori, docenti, innovatori, professionisti ed altre figure professionali che operano nel settore dell'informatica, della progettazione, della fabbricazione digitale e delle logiche connesse alla gestione sostenibile delle risorse idriche e ambientali.

PREMI

1500 € al primo team classificato
1000 € al secondo team classificato
500 € al terzo team classificato

ISCRIZIONE

La partecipazione è totalmente gratuita ed è aperta a massimo 50 partecipanti. Per partecipare alla Call è necessario compilare e trasmettere entro il 15 settembre 2026 il modulo online (<https://forms.gle/RE865LKUUeUBkkWw9>) allegando il curriculum in formato Europass e la copia di un documento di identità.

La partecipazione all'Hackathon consentirà agli studenti del corso triennale di Ingegneria Informatica e di Chimica Verde e Sostenibile dell'Università di Modena e Reggio Emilia (sede di Mantova) di ottenere 1 credito formativo inserendo l'attività nel piano di studi. È possibile ottenere 1 ulteriore credito formativo proseguendo l'attività con lo sviluppo di un prototipo della soluzione proposta. Rispetto a possibili analoghe opportunità con le altre Università in rete, si rimanda a successive verifiche che la Fondazione UniverMantova promuoverà con il Politecnico di Milano – Polo territoriale di Mantova e l'Università degli Studi di Brescia – Sede di Mantova.

PROGRAMMA

Mercoledì 23 settembre 2026

14.30 Accoglienza e formazione dei team
15.00 Introduzione al tema di progettazione
17.00 Avvio dei lavori da parte dei team

Giovedì 24 settembre 2026

9.00 Ripresa dei lavori da parte dei team
13.00 Pranzo
14.00 Ripresa dei lavori da parte dei team
17.30 Prima presentazione da parte dei team

Venerdì 25 settembre 2026

9.00 Ripresa dei lavori da parte dei team
13.00 Pranzo
14.00 Allestimento e ultimi ritocchi
16.00 Presentazione dei risultati
18.00 Premiazione dei vincitori all'interno della "Notte Europea della Ricerca 2026"

Organizzato da



Promosso da



In rete con



All'interno di



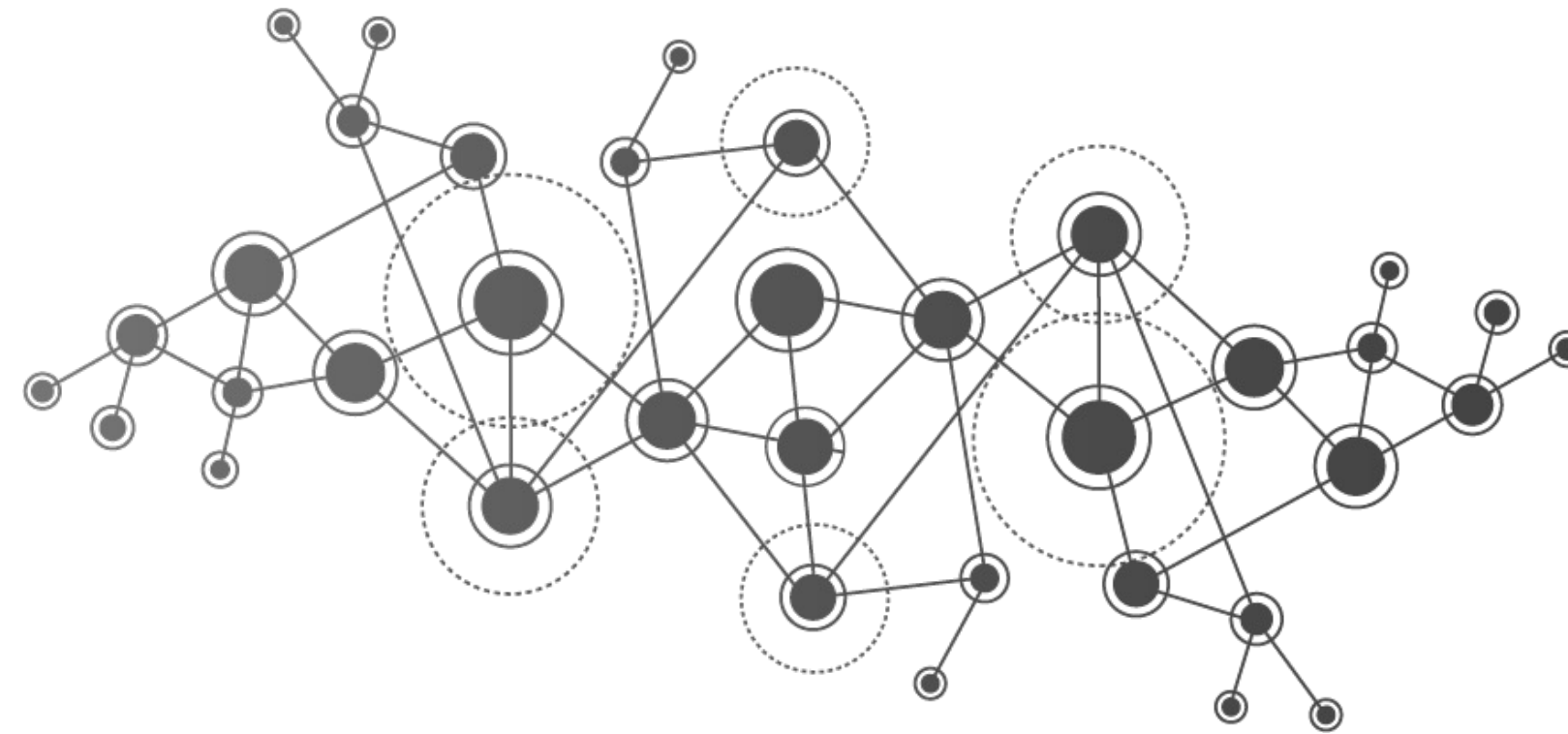
Con il sostegno di





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Lecture 1- Internet of Things Overview

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